# tree2go options reference guide

Every tree2go option, grouped as in the app, with the left panel section by section, the header menus, the `Right-click` menus and what you can do on the canvas. Each group folds. The **Appears when** column says what an option needs before it shows up, and single elements can also be set from their `Right-click` menu.

In the panel, sections and their subsections (marked ›) fold, and the open ones are remembered. `Collapse` closes everything, and `Expand` opens the sections and keeps their subsections as they were. Until a tree is open, every section but Tree and Attachments is grayed out.

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# Left panel

> The left panel sets the look of the whole tree at once. A change there applies to every element of that kind, so all the branches, all the tip labels or all the clade bars change together. This is where a figure gets its general style, section by section.

## Tree

Load or paste a tree (Newick and NEXUS are recognized from their content, whatever the file extension). A file with many trees (for example a BEAST or MrBayes posterior) opens as a tree set. If another program has rewritten a file's annotations in a way that breaks them, tree2go shows where and offers to repair the keys when loading it. It matches the braces and rebuilds the key names. Values that were removed from the file, such as rates, cannot be recovered.

Before reading a file with more than 10,000 trees, or any tree file larger than 150 MB, tree2go opens a window where you set a burn-in and how many trees to keep after it (a number or a percentage, evenly spaced or at random with a seed), so only the trees you keep are read. A sample of 10,000 trees is more than enough for summary trees. If you ask for more trees than remain after the burn-in, the window says so and does not load them. You can also read all the trees, and closing the window loads nothing. The Tree set part then shows, in a red box, how the trees were resampled, from how many, and with which burn-in, so that another burn-in is not applied by mistake. MrBayes clock trees carry their branch lengths in substitutions per site, with each tree's clock rate in its header `[&clockrate = …]`, tree2go divides each tree by its own clock rate, so the ages of the summary trees are in time units, as in the MrBayes consensus. Files larger than 150 MB are read in pieces: tree2go counts their trees first, then reads only the trees kept, so a posterior of hundreds of thousands of trees can be opened with a burn-in and a resample. Reading them all at once is limited by the memory of the browser. Every tree of a set is also kept in the browser, compressed, so the set comes back whole when the app is opened again, with its burn-in and summary. Only the last set is kept, and loading a single tree or `Clear session` removes it.

Several files of trees can be opened together, such as the `.t` files of the runs of one MrBayes analysis (`analysis.run1.t`, `analysis.run2.t` …), choosing them all at once in `Open tree` or dropping them together on the canvas. tree2go reads them as one set, numbering the runs in the order of their file names, and checks that they hold the same tips. The window always opens, whatever the number of trees: the burn-in leaves out the first trees of each run, the same number of trees is kept from each run, and the trees kept are combined. `Combine all after the burn-in` keeps every tree left after the burn-in. Each tree remembers its run, so the summary trees and the ASDSF can use only some runs (`Runs used`), and the MAP tree can find each tree in the `.p` file of its own run. Runs larger than 150 MB each are opened one at a time.

`Test performance` times tree2go on this computer with a tree of 20,000 tips, 1,000 trees of 1,000 tips and 10,000 trees of 100 tips (about 38 MB of extra data to download). Its button shows in this section when the app is opened with `?bench=manual` at the end of its address (for example `https://vr-daniel.github.io/tree2go/?bench=manual`). It measures opening the tree, zooming in, zooming out and moving it (each until the view is sharp and still), rerooting, ladderizing and exporting it (Newick, SVG, PDF), and reading each tree set and building its majority-rule consensus and MCC tree. Each result is shown next to the time on the computer tree2go was tuned on. `Copy results` copies them as text. Your work is kept and comes back as it was. The test needs an internet connection.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Burn-in` | Share of the first trees of the chain left out of every summary (of each run, when the trees come from several runs), applied with `Apply` (or `Enter`). Applying it rebuilds the summary tree on screen, so the tree and its age densities use the same trees. When the trees were already resampled while reading the file, the note on that resample stays, and a second note gives this second burn-in and the trees it leaves. | 0 to 99% | 0% | A file with several trees is open |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Runs used` [1] | The runs whose trees are used by the summary trees, support values, age densities and ASDSF, applied with `Apply` (or `Enter`). Applying it rebuilds the summary tree on screen. The runs are numbered in the order of their file names, shown when hovering the field, and the ASDSF of the runs used is given under it. | all<br>Run numbers (1,2,4)<br>A range (1-3) | all | The trees come from several runs |
| `Trees in time` | Whether the trees are read in time (dated with a clock, including trees dated by their tips with few or no tips at the present). Recognized automatically from the files. If tree2go gets it wrong, it can be set by hand. In dated trees, leaving it off misplaces everything that depends on node ages: summary trees then take the mean length of each branch instead of the ages of the nodes, and no longer line up with the age densities. | On<br>Off | Recognized from the trees | A file with several trees is open, with branch lengths and rooted |
| `Show tree number` | Shows one sampled tree of the file, by its position. | Number | 1 | A file with several trees is open |
| `Summary tree` | How the trees of the file are summarized into one. | Majority rule[2]<br>Strict consensus[3]<br>Maximum compatible[4]<br>Maximum clade credibility (MCC)[5]<br>Maximum a posteriori (MAP)[6] | Majority rule | A file with several trees is open, **MAP** needs the posterior of each tree in the file (as a `.p` or `.log` file in Attachments, one for each run used when the trees come from several runs) |
| `Clades in more than` | Clades must be found in more than this share of the trees to be kept. | 50 to 100% | 50% | `Summary tree` is **Majority rule** |
| `Node ages` | How the summary tree gets its node ages from all the trees. The median matches the center of the age densities drawn on the nodes. | Median<br>Mean | Median | `Summary tree` is not **MAP**, and the trees are dated and rooted |
| `Build summary tree` | Builds and shows the chosen summary tree. | | | A file with several trees is open |

[1] ASDSF (average standard deviation of split frequencies), as MrBayes reports it: for every split found in at least 10% of the trees of some run, the standard deviation of its frequency across the runs used, averaged over those splits. It needs at least two runs. MrBayes takes values below 0.01 as a sign that the runs converged on the same trees.
[2] Majority rule: keeps every clade found in more than the chosen share of the trees (more than half by default), and gives each one that share as its support.
[3] Strict consensus: keeps only the clades found in every tree. Where the trees disagree, the branches join in a polytomy.
[4] Maximum compatible: the majority-rule tree plus the less frequent clades that do not conflict with it, added from the most to the least frequent (allcompat in MrBayes).
[5] MCC (maximum clade credibility): the sampled tree whose clades have the highest product of their frequencies, as in TreeAnnotator. Each node gets the median or mean age of its clade in all the trees.
[6] MAP (maximum a posteriori): the sampled tree with the highest posterior probability, read from the trees file (BEAST) or from the attached log (`.log` of BEAST2, `.p` of MrBayes). With several runs, attach the `.p` file of each run (they can be

chosen together): each is matched with its run by the file name (`analysis.run2.p` with `analysis.run2.t`) or by the run number in it, and tree2go asks which run a file belongs to when its name does not say.

With unrooted trees (no clock), the branch lengths of the majority-rule, strict and maximum compatible trees are the mean length of each branch over the trees that contain it, as in MrBayes.

Every summary tree is annotated as TreeAnnotator does, using the trees that contain each clade. Each node gets its `posterior`, and the mean, median, 95% HPD and range of its height (in time trees), of its branch length and of every numeric annotation of the sampled trees. These annotations can come from BEAST, written as `[&rate=…]`, or from MrBayes, written as `[&B name value]`, such as the branch lengths of its relaxed-clock models. For an annotation such as `rate`, the summary tree gets `rate` (the mean), `rate_median`, `rate_95%_HPD` and `rate_range`. They can then be shown as node values, used to color branches or drawn as node bars.

### › Reading properties

For files with several trees: how tree2go keeps its copy of each tree (used to show another tree, pick the MCC or MAP tree, draw the tree cloud and export the set). They take effect with the next file opened. Supports, ages and summary trees are computed from the file at full precision.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Significant digits of branch lengths` [1] | Precision of the branch lengths in the copy kept of each tree. | 1 to 10 | 6 | |
| `Keep the trees in a compact binary form` [2] | Keeps each tree as numbers (topology, branch lengths, tip numbers and supports) instead of text, which takes about half the memory. | On<br>Off | On | |

[1] Fewer than 6 digits can round short branches, or make them zero, while more digits keep more precision but take more memory and time with many trees.
[2] Without it each tree is kept as text, as in earlier versions. Networks are always kept as text.

## Layout

The shape of the tree and its overall size.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Shape` [1] | Overall form of the tree. | Rectangular<br>Circular<br>Fan<br>Unrooted | Rectangular | |
| `Orientation` | Horizontal, with the tips on the right, vertical, with the tips on top and the root at the bottom, or custom, turned by any angle. The time axis, the scale bar and the clade names turn with the tree, legends and the title do not. Split into pages works on horizontal trees only. | Horizontal<br>Vertical<br>Custom | Horizontal | `Shape` is **Rectangular** |
| `Angle` | Turns the tree clockwise, where 0° is horizontal and 270° is vertical with the tips on top. At other angles the text turns with the tree, and pages and alignment guides are off, as on circular trees. | 0 to 359° | 0° | `Shape` is **Rectangular**, `Orientation` is **Custom** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Scale bar on a vertical tree` | Turns with the tree, vertical like the branches it measures, or stays horizontal under the root. Both measure the same, since the figure keeps one scale in both directions. The time axis always turns with the tree. | Turns with the tree<br>Horizontal | Turns with the tree | `Shape` is **Rectangular**, `Orientation` is **Vertical**, `Show scale bar` is **On**, `Show time axis (dated trees)` is **Off** |
| `Text on a vertical tree` | Vertical: every text reads from bottom to top. Horizontal: the tip names read at 45 degrees and the values of nodes and branches across, beside their branch. The number of the scale bar always follows its bar, and the texts of the side plots (category names, heatmap titles, bar values and bar scale) always stay upright. | Vertical<br>Horizontal | Vertical | `Shape` is **Rectangular**, `Orientation` is **Vertical** |
| `Use branch lengths` | Draws branches proportional to their lengths, when **Off** the tree is a cladogram. | On<br>Off | On | |
| `Order nodes` [2] | Sorts every node by clade size, once, for the whole tree. | Increasing<br>Decreasing | | |
| `Rename tips from a file…` | Renames many tips at once from a two-column file, with the current name first and the new name second. Column headers are optional. A renamed tip keeps its original name underneath, so it stays linked to its tables, alignments, plots and maps. The new name is shown everywhere. The same holds for a tip renamed from its `Right-click` menu. When a tree with renamed tips is exported as Newick or NEXUS, tree2go asks whether to write the new names, as shown in the figure, or the original ones, as in the data. | File | | |
| `Width` | Horizontal size of the tree, without labels. | 80 to 4000 px | 700 px | **Rectangular** layout |
| `Height` | Vertical distance between neighboring tips. | 3 to 80 px | 18 px | **Rectangular** layout |
| `Radius` | | 40 to 3000 px | 320 px | **Circular**, **Fan** or **Unrooted** layout |
| `Inner radius (center gap)` | Empty circle left at the center. | 0 to 1500 px | 0 px | `Shape` is **Circular** or **Fan** |
| `Total angle` | How much of the circle the tree fills. | 20 to 360° | 360° | `Shape` is **Circular** |
| `Fan opening` | Angle covered by a fan tree. | 20 to 359° | 180° | `Shape` is **Fan** |
| `Rotation` | Turns the whole circular, fan or unrooted tree. | −180 to 180° | 0° | **Circular**, **Fan** or **Unrooted** layout |
| `Root branch` | Short branch drawn before the root, which can hold its values. | 0 to 80 px | 0 px | `Shape` is not **Unrooted** |

[1] Locked while the page has images (maps do not count).
[2] Affects the whole tree.

## Tree cloud

Draws the trees of a dated posterior sample (after the burn-in) faintly behind the current tree, using its tip order and time axis. The cloud looks dense where the trees agree and spread out where they differ. It needs a file with several dated trees and a rectangular tree. Its branches follow the angle and the rounded corners of the tree, and it turns with it.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show the trees of the sample` | Draws the cloud of trees with the tree shown. | On<br>Off | Off | A dated tree set is loaded |
| `Trees drawn` | As a number of trees or as a percentage of the sample. Drawing more trees can slow the view down considerably, since every tree adds lines for the browser to draw. | A number<br>A percentage | A number | `Show the trees of the sample` is **On** |
| `Number of trees` | | 1 to 100000 | 300 | `Show the trees of the sample` is **On**, `Trees drawn` is **A number** |
| `Percentage of the trees` | | 1 to 100% | 10% | `Show the trees of the sample` is **On**, `Trees drawn` is **A percentage** |
| `Which trees` | Evenly spaced along the sample, or a random sample that the seed makes repeatable. | Evenly spaced<br>At random | Evenly spaced | `Show the trees of the sample` is **On** |
| `Seed` | The same seed always draws the same trees. | 0 to 999999 | 123 | `Show the trees of the sample` is **On**, `Which trees` is **At random** |
| `Color` | | Color | #2F5D8C | `Show the trees of the sample` is **On** |
| `Opacity of each tree` | Where many trees overlap, the cloud gets darker. | 1 to 100% | 6% | `Show the trees of the sample` is **On** |
| `Line width` | | 0.1 to 10 px | 0.8 px | `Show the trees of the sample` is **On** |
| `Place` | | Behind the tree<br>In front | Behind the tree | `Show the trees of the sample` is **On** |

## Branches

Width, color and line style of every branch (a value given to a branch from its `Right-click` menu always wins over the general panel).

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Line width` | | 1 to 20 px | 1 px | |
| `Branch color` | | Color | #1F1F1F | |
| `Line ends` | Shape of the ends of every line. | Round<br>Square | Round | |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Branch angle` | Angle at which branches leave their node. At 90° you get the classic square tree, and lower values slant them. | 45 to 90° | 90° | **Rectangular** layout |
| `Rounded corners` | Rounds the bend where a branch turns toward its child. | 0 to 30 px | 0 px | **Rectangular** layout |
| `Line style` | | Solid<br>Dashed<br>Dotted | Solid | |
| `Dash length (also fossils)` | Used by dashed branches and by grafted fossils. | 1 to 30 px | 5 px | |
| `Gap between dashes or dots` | | 1 to 30 px | 3 px | |

## › Color by data

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Value` [1] | The tree annotation or table column that sets the colors. They all come in one list, with the branch length (raw) first, then every tree annotation and every table column (with the table name). | List | None | |
| `Palette` | The scale colors, either your own or a standard palette such as Viridis or Inferno. Those marked ✓ stay distinguishable with the common forms of color blindness. | Custom colors<br>tree2go<br>Okabe-Ito ✓<br>Paul Tol bright ✓<br>Viridis ✓<br>Inferno ✓<br>Magma ✓<br>Plasma ✓<br>Cividis ✓<br>Turbo<br>Spectral<br>Blue to red ✓ | Custom colors | `Value` is set |
| `Reverse the colors` | Runs the palette the other way, low values taking the colors of the high ones. | On<br>Off | Off | `Value` is set |
| `Hue shift` | Turns the hue of every color of the palette. | −180 to 180° | 0° | `Value` is set |
| `Low values` | | Color | #2C7BB6 | `Value` is set, `Palette` is **Custom colors** |
| `Use a middle color` | Adds a third color for values in the middle of the range. | On<br>Off | On | `Value` is set, `Palette` is **Custom colors** |
| `Middle values` | | Color | #FFFFBF | `Value` is set, `Use a middle color` is **On**, `Palette` is **Custom colors** |
| `High values` | | Color | #D7191C | `Value` is set, `Palette` is **Custom colors** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Logarithmic scale (log₁₀)` | Uses the base-10 logarithm of the values for the colors, so each tenfold step gets the same share. The legend shows the values themselves. | On<br>Off | Off | `Value` is set |
| `Lowest value of the scale` | Fixes the start of the scale, empty uses the lowest value found. | $-10^{12}$ to $10^{12}$ | | `Value` is set |
| `Highest value of the scale` | Fixes the end of the scale, empty uses the highest value found. | $-10^{12}$ to $10^{12}$ | | `Value` is set |
| `Show a color legend` | | On<br>Off | On | `Value` is set |

[1] Color the branches by a value read from the tree, such as a rate or a posterior annotation, or from a table in Attachments with tips and internal nodes named as in the tree.

### › Width by data

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Value` [1] | The tree annotation or table column that sets the widths. They all come in one list, with the branch length (raw) first, then every tree annotation and every table column (with the table name). | List | None | |
| `Thinnest` | Width of the branch with the lowest value. | 0.1 to 10 px | 0.5 px | `Value` is set |
| `Thickest` | Width of the branch with the highest value. | 0.5 to 20 px | 5 px | `Value` is set |
| `Logarithmic scale (log₁₀)` | Uses the base-10 logarithm of the values for the widths, so each tenfold step gets the same share. The legend shows the values themselves. | On<br>Off | Off | `Value` is set |
| `Show legend` | | On<br>Off | On | `Value` is set |

[1] Make the branches thicker where a value is higher, for example a rate or a posterior probability, read from the tree or from an attached table.

## Branch values

Labels with the length or annotations for each branch.

### › Values on the branches

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show a value on each branch` | Writes a number on every branch. | On<br>Off | Off | |
| `Value` | One list with the branch length (raw) first, then every tree annotation and every table column (with the table name). | List | Branch length | `Show a value on each branch` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Decimals` | | 0, 1, 2, 3, 4, 5, 6 | 3 | `Show a value on each branch` is **On** |
| `Text size` | | 1 to 100 px | 8 px | `Show a value on each branch` is **On** |
| `Color of the values` | One color for all, the color of each branch (its color by data, a clade color or the line color), or a color from a parameter of its own, whatever the color of the branches. | One color<br>Like their branch<br>By data | One color | `Show a value on each branch` is **On** |
| `Colored by` | The parameter that gives the colors, such as branch length (raw), a tree annotation or a table column. | List | Branch length (raw) | `Show a value on each branch` is **On**, `Color of the values` is **By data** |
| `Palette` | Scale colors. Those marked ✓ stay distinguishable with the common forms of color blindness. | Custom colors<br>tree2go<br>Okabe-Ito ✓<br>Paul Tol bright ✓<br>Viridis ✓<br>Inferno ✓<br>Magma ✓<br>Plasma ✓<br>Cividis ✓<br>Turbo<br>Spectral<br>Blue to red ✓ | Viridis | `Show a value on each branch` is **On**, `Color of the values` is **By data** |
| `Low values` | | Color | #2C7BB6 | `Show a value on each branch` is **On**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Use a middle color` | Adds a third color for values in the middle of the range. | On<br>Off | On | `Show a value on each branch` is **On**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Middle values` | | Color | #FFFFBF | `Show a value on each branch` is **On**, `Color of the values` is **By data**, `Palette` is **Custom colors**, `Use a middle color` is **On** |
| `High values` | | Color | #D7191C | `Show a value on each branch` is **On**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Reverse the colors` | | On<br>Off | Off | `Show a value on each branch` is **On**, `Color of the values` is **By data** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show a color legend` | Adds a legend with the colors and the range of the values. | On<br>Off | On | `Show a value on each branch` is **On**, `Color of the values` is **By data** |
| `Color` | | Color | ■ #666666 | `Show a value on each branch` is **On**, `Color of the values` is **One color** |
| `Position on branch` | Where along the branch the value is written. | Left<br>Center<br>Right | Center | `Show a value on each branch` is **On** |
| `Margin from branch end` | Distance from the end of the branch, for left and right positions. | −2000 to 2000 px | 0 px | `Show a value on each branch` is **On**, `Position on branch` is not **Center** |
| `Vertical offset` | Moves the value up (positive) or down (negative). | −500 to 500 px | 2 px | `Show a value on each branch` is **On** |

## Nodes

Dots on tips and internal nodes.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Dots on tips` | | On<br>Off | Off | |
| `Tip dot shape` | Shape of the dots on the tips. Any dot can take its own shape from its `Right-click` menu. | Circle<br>Square<br>Triangle<br>Diamond<br>Star | Circle | `Dots on tips` is **On** |
| `Tip dot size` | | 0.5 to 12 px | 3 px | `Dots on tips` is **On** |
| `Dots on internal nodes` | | On<br>Off | Off | |
| `Node dot shape` | Shape of the dots on the internal nodes. | Circle<br>Square<br>Triangle<br>Diamond<br>Star | Circle | `Dots on internal nodes` is **On** |
| `Node dot size` | | 0.5 to 12 px | 3 px | `Dots on internal nodes` is **On** |
| `Dot opacity` | | 5 to 100% | 100% | `Dots on tips` is **On**, `Dots on internal nodes` is **On** |
| `Fill dots` | | On<br>Off | On | `Dots on tips` is **On**, `Dots on internal nodes` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Dot color` | | Color | ■ #1F1F1F | `Dots on tips` is **On**, `Dots on internal nodes` is **On**, `Fill dots` is **On** |
| `Dot border` | | Solid<br>Dashed<br>Dotted<br>None | None | `Dots on tips` is **On**, `Dots on internal nodes` is **On** |
| `Border color` | | Color | ■ #1F1F1F | `Dots on tips` is **On**, `Dots on internal nodes` is **On**, `Dot border` is not **None** |
| `Border width` | | 0.05 to 50 px | 1 px | `Dots on tips` is **On**, `Dots on internal nodes` is **On**, `Dot border` is not **None** |
| `Number of dashes or dots` | How many dashes or dots go around the dot. | 1 to 60 | 12 | `Dots on tips` is **On**, `Dots on internal nodes` is **On**, `Dot border` is **Dashed** or **Dotted** |

## › Dots by data

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Value` [1] | The tree annotation or table column that sets the dots. They all come in one list, with the branch length (raw) first, then every tree annotation and every table column (with the table name). | List | None | |
| `Palette` | The scale colors, either your own or a standard palette such as Viridis or Inferno. Those marked ✓ stay distinguishable with the common forms of color blindness. | Custom colors<br>tree2go<br>Okabe-Ito ✓<br>Paul Tol bright ✓<br>Viridis ✓<br>Inferno ✓<br>Magma ✓<br>Plasma ✓<br>Cividis ✓<br>Turbo<br>Spectral<br>Blue to red ✓ | Custom colors | `Value` is set, `Color by the value` is **On** |
| `Reverse the colors` | Runs the palette the other way, low values taking the colors of the high ones. | On<br>Off | Off | `Value` is set, `Color by the value` is **On** |
| `Hue shift` | Turns the hue of every color of the palette. | −180 to 180° | 0° | `Value` is set, `Color by the value` is **On** |
| `Color by the value` | Colors each dot along a scale from its value. | On<br>Off | On | `Value` is set |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Low values` | | Color | #2C7BB6 | `Value` is set, `Color by the value` is **On**, `Palette` is **Custom colors** |
| `High values` | | Color | #D7191C | `Value` is set, `Color by the value` is **On**, `Palette` is **Custom colors** |
| `Size by the value` | Makes each dot bigger as its value grows. | On<br>Off | Off | `Value` is set |
| `Smallest` | Size of the dot with the lowest value. | 0.5 to 10 px | 2 px | `Value` is set, `Size by the value` is **On** |
| `Largest` | Size of the dot with the highest value. | 1 to 20 px | 7 px | `Value` is set, `Size by the value` is **On** |
| `Show legend` | | On<br>Off | On | `Value` is set |

[1] Give the dots a color or a size from a value, for example the posterior or a rate read from the tree, or a column of a table (nodes with a value always get a dot).

## Node values

Support values, other values read from the tree (such as ages or probabilities), and bars on the nodes (HPD intervals or age densities from a tree set).

### › Support values

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show as` | How support values are shown on the nodes. | Hidden<br>Number<br>Circles | Hidden | |
| `Values from` [1] | Where support values come from. | Input tree<br>SIMMAP<br>Trees file | Input tree | `Show as` is not **Hidden** |
| `Decimals` | Decimals of the numbers written, or as they are written in the file. | As written<br>0, 1, 2, 3 | As written | `Show as` is **Number** |
| `Size` | | 1 to 100 px | 9 px | `Show as` is **Number** or **Circles**, `Which values` is **Above a threshold** |
| `Color of the values` | One color for all, or the color of the branch leading to each node, or a color from a parameter of its own, whatever the color of the branches. | One color<br>Like their branch<br>By data | One color | `Show as` is not **Hidden**, `Which values` is **Above a threshold** |
| `Colored by` | The parameter that gives the colors, such as branch length (raw), a tree annotation or a table column. | List | Branch length (raw) | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Palette` | Scale colors. Those marked ✓ stay distinguishable with the common forms of color blindness. | Custom colors<br>tree2go<br>Okabe-Ito ✓<br>Paul Tol bright ✓<br>Viridis ✓<br>Inferno ✓<br>Magma ✓<br>Plasma ✓<br>Cividis ✓<br>Turbo<br>Spectral<br>Blue to red ✓ | Viridis | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data** |
| `Low values` | | Color | #2C7BB6 | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Use a middle color` | Adds a third color for values in the middle of the range. | On<br>Off | On | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Middle values` | | Color | #FFFFBF | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data**, `Palette` is **Custom colors**, `Use a middle color` is **On** |
| `High values` | | Color | #D7191C | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Reverse the colors` | | On<br>Off | Off | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data** |
| `Show a color legend` | Adds a legend with the colors and the range of the values. | On<br>Off | On | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **By data** |
| `Color` | | Color | #6A6A6A | `Show as` is not **Hidden**, `Which values` is **Above a threshold**, `Color of the values` is **One color** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Bigger circles for higher values` | Circle size grows with the support. | On<br>Off | Off | `Show as` is **Circles**, `Which values` is **Above a threshold** |
| `Which values` | Show the values above a threshold, or style them by ranges. | Above a threshold<br>By categories | Above a threshold | `Show as` is not **Hidden** |
| `Only when support ≥` | Values below this are not shown. | 0 to 100 | 0 | `Show as` is not **Hidden**, `Which values` is **Above a threshold** |
| `Ranges` | Each range of support gets its own look. By default, higher ranges get bigger circles (3, 3.5 and 4 px): give every range the same size to change only the color. | One card per range (from, to, fill, size, border) | | `Show as` is not **Hidden**, `Which values` is **By categories** |
| `Show category legend` | | On<br>Off | On | `Show as` is not **Hidden**, `Which values` is **By categories** |
| `Position` | Side of the node, or the middle of its branch. | Left<br>Centered on the branch<br>Right | Left | `Show as` is **Number** |
| `Distance from the node` | Space between the node and the value. | −2000 to 2000 px | 3 px | `Show as` is **Number**, `Position` is not **Centered on the branch** |
| `Vertical offset` | Moves the value up (positive) or down (negative). | −500 to 500 px | 2 px | `Show as` is **Number** |

[1] Input tree: the labels of its internal nodes. Trees file: the percentage of trees that contain the same clade, as in bootstrap or posterior support.

## › Other node labels

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show` [1] | Value written next to each node, such as names, annotations or tree set summaries. | List | | |
| `Show as` | Writes each value, or draws a circle whose color and size can follow it. | Number<br>Circles | Number | `Show` is set |
| `Color by the value` | Colors each circle along a scale from its value. | On<br>Off | On | `Show` is set, `Show as` is **Circles** |
| `Low values` | | Color | #2C7BB6 | `Show` is set, `Show as` is **Circles**, `Color by the value` is **On** |
| `High values` | | Color | #D7191C | `Show` is set, `Show as` is **Circles**, `Color by the value` is **On** |
| `Size by the value` | Makes each circle bigger as its value grows. | On<br>Off | Off | `Show` is set, `Show as` is **Circles** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Circle size` | | 0.5 to 15 px | 3.5 px | `Show` is set, `Show as` is **Circles**, `Size by the value` is **Off** |
| `Smallest` | Size of the circle with the lowest value. | 0.5 to 10 px | 2 px | `Show` is set, `Show as` is **Circles**, `Size by the value` is **On** |
| `Largest` | Size of the circle with the highest value. | 1 to 20 px | 7 px | `Show` is set, `Show as` is **Circles**, `Size by the value` is **On** |
| `Show legend` | | On<br>Off | On | `Show` is set, `Show as` is **Circles** |
| `Decimals` | | 0, 1, 2, 3, 4, 5, 6 | 2 | `Show` is set, `Show as` is **Number** |
| `Text size` | | 1 to 100 px | 9 px | `Show` is set, `Stack with the support values` is not **After the support (a / b)**, `Show as` is **Number** |
| `Color of the values` | One color for all, or the color of the branch leading to each node, or a color from a parameter of its own, whatever the color of the branches. | One color<br>Like their branch<br>By data | One color | `Show` is set, `Show as` is **Number** |
| `Colored by` | The parameter that gives the colors, such as branch length (raw), a tree annotation or a table column. | List | Branch length (raw) | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data** |
| `Palette` | Scale colors. Those marked ✓ stay distinguishable with the common forms of color blindness. | Custom colors<br>tree2go<br>Okabe-Ito ✓<br>Paul Tol bright ✓<br>Viridis ✓<br>Inferno ✓<br>Magma ✓<br>Plasma ✓<br>Cividis ✓<br>Turbo<br>Spectral<br>Blue to red ✓ | Viridis | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data** |
| `Low values` | | Color | #2C7BB6 | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Use a middle color` | Adds a third color for values in the middle of the range. | On<br>Off | On | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data**, `Palette` is **Custom colors** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Middle values` | | Color | #FFFFBF | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data**, `Palette` is **Custom colors**, `Use a middle color` is **On** |
| `High values` | | Color | #D7191C | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data**, `Palette` is **Custom colors** |
| `Reverse the colors` | | On<br>Off | Off | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data** |
| `Show a color legend` | Adds a legend with the colors and the range of the values. | On<br>Off | On | `Show` is set, `Show as` is **Number**, `Color of the values` is **By data** |
| `Color` | | Color | #6A6A6A | `Show` is set, `Stack with the support values` is not **After the support (a / b)**, `Show as` is **Number** or `Color by the value` is **Off**, `Color of the values` is **One color** |
| `Position` | Side of the node, or the middle of its branch. | Left<br>Centered on the branch<br>Right | Left | `Show` is set, `Stack with the support values` is not **After the support (a / b)**, `Show as` is **Number** |
| `Distance from the node` | Space between the node and the label. | −2000 to 2000 px | 3 px | `Show` is set, `Position` is not **Centered on the branch**, `Stack with the support values` is not **After the support (a / b)**, `Show as` is **Number** |
| `Vertical offset` | Moves the label up (positive) or down (negative). | −500 to 500 px | 2 px | `Show` is set, `Stack with the support values` is not **After the support (a / b)**, `Show as` is **Number** |
| `Stack with the support values` | Writes these labels together with the support values. | No<br>Below the branch<br>After the support (a / b) | No | `Show` is set, `Show as` is **Number** |

[1] Load a SIMMAP or a tree with annotations to write values next to the nodes.

## › Node bars

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Draw` [1] | Uncertainty drawn on each node, either an interval from the tree or the ages from a trees file. | List | | |
| `Show the ages as` | Full distribution of the ages, or only its HPD interval. | Violin<br>HPD bar | Violin | `Draw` is set |
| `Height` | Thickness of the bar, or height of the violin on each side. | 1 to 40 px | 6 px | `Draw` is set |
| `Color` | | Color | #5B8DEF | `Draw` is set |
| `Opacity` | | 5 to 100% | 45% | `Draw` is set |
| `Border` | | Solid<br>Dashed<br>Dotted<br>None | None | `Draw` is set |
| `Border color` | | Color | #2F5D8C | `Draw` is set, `Border` is not **None** |
| `Border width` | | 0.05 to 50 px | 0.8 px | `Draw` is set, `Border` is not **None** |
| `Mirror the density on both sides of the branch` | Draws the violin above and below the branch. | On<br>Off | On | `Draw` is set, `Show the ages as` is not **HPD bar** |
| `Smoothing` | Bandwidth of the density. Lower values keep more detail and higher values give a smoother curve. | 0.2 to 3× | 1× | `Draw` is set, `Show the ages as` is not **HPD bar** |
| `Show the HPD inside the violin` | Draws the HPD interval as a thin bar over the violin. | On<br>Off | Off | `Draw` is set, `Show the ages as` is not **HPD bar** |
| `HPD interval` | Share of the ages the interval holds. | 90%, 95%, 99% | 95% | `Draw` is set, `Show the ages as` is **HPD bar**, `Show the HPD inside the violin` is **On** |
| `HPD thickness` | | 0.5 to 20 px | 3 px | `Draw` is set, `Show the ages as` is not **HPD bar**, `Show the HPD inside the violin` is **On** |
| `HPD color` | | Color | #1F1F1F | `Draw` is set, `Show the ages as` is not **HPD bar**, `Show the HPD inside the violin` is **On** |
| `Mark the median` | A short line at the median age. | On<br>Off | Off | `Draw` is set |
| `Median line width` | | 0.1 to 20 px | 1.5 px | `Draw` is set, `Mark the median` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Median line` | | Solid<br>Dashed<br>Dotted | Solid | `Draw` is set, `Mark the median` is **On** |
| `Median color` | | Color | ■ #1F1F1F | `Draw` is set, `Mark the median` is **On** |
| `Mark the mean` | A short line at the mean age, next to or instead of the median. | On<br>Off | Off | `Draw` is set |
| `Mean line width` | | 0.1 to 20 px | 1.2 px | `Draw` is set, `Mark the mean` is **On** |
| `Mean line` | | Solid<br>Dashed<br>Dotted | Dashed | `Draw` is set, `Mark the mean` is **On** |
| `Mean color` | | Color | ■ #C8553D | `Draw` is set, `Mark the mean` is **On** |

[1] Ranges from the tree annotations, such as height_95%_HPD, or age densities from a file with several dated trees.

### › Pie charts

Pie charts on the tips and on named internal nodes, read from a file. The file can have one pie per line as *Name | {A:0.3, B:0.7}*, as in SIMMAP files, or be a table with the names in the first column and one column per category. Internal nodes go by their names in the tree, as for the other attachments. Each pie adds up to 1, and the tip names move out to make room for it, as they also do for the SIMMAP pies on the tips.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Pie charts from a file…` | Reads the file, gives each category a color of the palette and shows the pies. | | | A tree is loaded |
| `Show the pies` | | On<br>Off | On | A pie file is loaded |
| `Place` | On the node, or in the middle of the branch that leads to it. | On the node<br>Middle of the branch | On the node | `Show the pies` is **On** |
| `Radius` | | 1 to 50 px | 5 px | `Show the pies` is **On** |
| `Border width` | | 0 to 5 px | 0.5 px | `Show the pies` is **On** |
| `Border color` | | Color | ■ #333333 | `Show the pies` is **On**, `Border width` is above 0 |
| `Show legend` | | On<br>Off | On | `Show the pies` is **On** |
| `Legend title` | | Text | | `Show the pies` is **On**, `Show legend` is **On** |
| `Category colors` | One color for each category of the file. | Color | Palette colors | A pie file is loaded |

## Tip labels

The names at the tips, with their font, size, style, color, and where they come from.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show tip names` | | On<br>Off | On | |
| `Show` | What the tips show, either their names or any value from one list (branch length, age, the annotations of the tips such as height or rate, and the columns of the tables). | List | The names in the tree | `Show tip names` is **On** |
| `Keep the name, with the value after it` | | On<br>Off | Off | `Show` is not **The names in the tree** |
| `Decimals` | | 0 to 4 | 2 | `Show` is not **The names in the tree** |
| `Font size` | | 1 to 200 px | 12 px | |
| `Typeface` | | Helvetica / Arial<br>Verdana<br>Trebuchet<br>Georgia<br>Times<br>Courier | Helvetica / Arial | |
| `Italic` | | On<br>Off | On | |
| `Bold` | | On<br>Off | Off | |
| `Keep taxonomic qualifiers upright (cf. / aff. / gr. / sp.)` | Writes qualifiers upright inside italic names, as the codes of nomenclature ask. | On<br>Off | On | `Italic` is **On** |
| `Label color` | | Color | ■ #1F1F1F | |
| `Labels take their clade color` [1] | Tip names follow the color of their clade. | On<br>Off | On | |
| `Distance from branch` | | 0 to 60 px | 5 px | |
| `Show underscores as spaces` | Shows *Homo_sapiens* as *Homo sapiens*. | On<br>Off | On | |
| `Label direction on circular, fan and unrooted trees` | Easy to read flips the labels on the left half, all outward keeps them pointing away from the center. | Easy to read<br>All outward | Easy to read | **Circular**, **Fan** or **Unrooted** layout |
| `Align labels to the right` | Lines up every name in one column, with guides back to the tips. | On<br>Off | Off | `Shape` is not **Unrooted** |
| `Alignment guide` | Line from each tip to its aligned name. | Dotted<br>Solid<br>None | Dotted | `Shape` is not **Unrooted**, `Align labels to the right` is **On** |

[1] Tip names use the color given to the branches of their clade.

# Clades

Highlights behind clades, collapsed clades and clade name bars.

## › Highlights

Add a highlight from the `Right-click` menu of a branch or a clade (these options set how every highlight looks).

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Highlight opacity` | | 0.05 to 1 | 0.55 | |
| `Starts at` [1] | Where the colored background of a clade begins. | Root<br>Stem<br>Node<br>Crown | Node | |
| `Ends at` | Where the colored background of a clade ends. | The tree edge<br>The clade labels | The tree edge | |
| `Gradient` | Fades the background toward one end. | None<br>Fade out<br>Fade in | None | |
| `Corner rounding` | | 0 to 20 px | 0 px | **Rectangular** layout |
| `Highlight clades from a file…` | One group per line, as for the clade names, to highlight each group in the next color of the palette (Page settings). Each highlight can then be changed with `Right-click`, like any other. | | | A tree is loaded |

[1] The whole clade, from its own node.

## › Collapsed clades

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Triangle height` | Rows a collapsed clade takes. | 1 to 10 rows | 2 rows | |
| `Fill collapsed triangles` | | On<br>Off | On | |
| `Show tip count on collapsed clades` | Writes how many tips each collapsed clade holds. | On<br>Off | On | |

## › Clade name bars

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show the names next to the bars` | Off, only the bars are drawn. | On<br>Off | On | |
| `Names on circular and fan trees` | Names pointing outward, or written along the arc. | Radial<br>Curved along the arc | Radial | **Circular** or **Fan** layout, `Show the names next to the bars` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Position along the arc` | Where a curved name sits along its bar. | Start<br>Center<br>End | Center | **Circular** or **Fan** layout, `Show the names next to the bars` is **On**, `Names on circular and fan trees` is **Curved along the arc** |
| `Clade bar width` | | 1 to 12 px | 3 px | |
| `Line up the clade bars in one column` [1] | Puts every bar at the same distance. | On<br>Off | On | |
| `Clade bar distance from the previous element` | Space between the clade name bars and the layer before it in `Layers`. Changing it moves the clade name bars and every layer after it together, as one block, and making it smaller brings them back the same way. | −200 to 200 px | 6 px | |
| `Clade name distance from bar` | | 0 to 60 px | 6 px | |
| `Clade name size` | | 1 to 200 px | 13 px | |
| `Clade names in bold` | | On<br>Off | On | |
| `Clade names in italic` | | On<br>Off | On | |
| `Name clades from a file…` | Adds many clade name bars at once from a file with one group per line, like *Primates={Homo sapiens, Pan troglodytes}*. One name marks that tip, two or more names their most recent common ancestor. | File | | |

[1] When off, each bar sits right after the labels of its own clade.

## Fossils

Extinct tips and grafted fossils. Mark a tip as extinct, or add a fossil lineage to a branch, from its `Right-click` menu.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Tips that end before the present are extinct` [1] | In dated trees, marks tips that do not reach the present as fossils. | On<br>Off | On | |
| `Tolerance (% of the tree height)` | Tips this close to the present still count as living. At 0, only tips at the present count as living. For very recent fossils, type a smaller value in the box (0.001%, for example). | 0 to 5% (smaller values can be typed) | 0.5% | `Tips that end before the present are extinct` is **On** |
| `Add † after the names of extinct tips and fossils` | | On<br>Off | On | |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Branches of extinct tips` [2] | Line style of extinct lineages. | Same<br>Dashed<br>Dotted | Dashed | |

[1] In dated trees, when the youngest tip is above 0, as in a fossil-only tree, every tip is extinct ( `Right-click` a tip to change it by hand).
[2] Grafted fossil lineages are always drawn dashed, using the dash length and gap from Branches.

## › Sampled ancestors

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show tips on zero-length branches as sampled ancestors` [1] | Draws sampled ancestors of fossilized birth-death trees as dots on their branch. A tree counts as drawn in time when several living tips end together at the present, or, for trees of fossils only, when it has tips on branches of exactly zero, the node heights of BEAST, or the names BEAST gives the trees of its posterior sample ( `STATE_0` , `STATE_1000` ...). In a tree of substitutions a zero-length tip is just a very short branch, so the option is greyed out. | On<br>Off | On | The tree is drawn in time |
| `Dot size` | | 0.5 to 10 px | 3 px | `Show tips on zero-length branches as sampled ancestors` is **On**, on a tree drawn in time |
| `Dot color` | | Color | ■ #1F1F1F | `Show tips on zero-length branches as sampled ancestors` is **On**, on a tree drawn in time |
| `Names` | Side of the branch where their names go. | Above the branch<br>Below the branch | Above the branch | `Show tips on zero-length branches as sampled ancestors` is **On**, on a tree drawn in time |
| `Align the names with their dot` | | Left<br>Center<br>Right | Right | `Show tips on zero-length branches as sampled ancestors` is **On**, on a tree drawn in time |
| `Distance from the branch` [2] | Space between the branch and the names. | −50 to 200 px | 0 px | `Show tips on zero-length branches as sampled ancestors` is **On**, on a tree drawn in time |

[1] Fossilized birth-death analyses write sampled ancestors as tips on branches of length zero. Each one is drawn as a dot on the branch of its descendants, with its name next to it.
[2] `Right-click` the name of one sampled ancestor to move only that one, for example when it overlaps another name.

## Phylogenetic networks

Networks written in extended Newick (eNewick), as PhyloNet, SNaQ or Dendroscope write them. A node with two parents appears twice with the same label: `#H` (hybridization), `#LGT` (lateral gene transfer) or `#R` (recombination). The occurrence with a subtree, like `(6)#H7`, is the hybrid node, the one that receives, and the bare `#H7` hangs from the donor lineage. A hybrid node has two parents. The major edge brings most of its genome (γ above 0.5) and the minor edge brings the rest, as in introgression. Without the minor edges, what remains is the major tree, drawn with solid lines. A network cannot be rerooted on or below a hybrid node, since both of its parents must stay above it. Gamma (γ), the last field of `name:length:support:γ`, is the share of the genome inherited through each line. Click a line or its value to select it, and `Right-click` it to change only that one. Save Newick writes the network back in eNewick.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show network number` | Shows one of the networks of a file by its number, such as the runs listed by SNaQ, in the order of their score. | Number | 1 | The file holds several networks |
| `Show the reticulations` | Draws each reticulation as a dashed line from the donor lineage to the hybrid node. | On<br>Off | On | A network is open |
| `Hybridization (#H)` | Color of the #H reticulations. | Color | #C8553D | `Show the reticulations` is **On**, the network has #H reticulations |
| `Lateral gene transfer (#LGT)` | Color of the #LGT reticulations. | Color | #2F5D8C | `Show the reticulations` is **On**, the network has #LGT reticulations |
| `Recombination (#R)` | Color of the #R reticulations. | Color | #7A4E9A | `Show the reticulations` is **On**, the network has #R reticulations |
| `Width from γ` | Thicker lines for a larger share of the genome inherited through them. | On<br>Off | On | `Show the reticulations` is **On**, the network has γ values |
| `Width` | | 0.1 to 20 px | 1.5 px | `Show the reticulations` is **On**, `Width from γ` is **Off** |
| `Thinnest (γ = 0)` | Width of a line with γ = 0. Each line takes a width between this and the thickest, in proportion to its γ. | 0.1 to 20 px | 0.6 px | `Show the reticulations` is **On**, `Width from γ` is **On** |
| `Thickest (γ = 1)` | Width of a line with γ = 1. With γ = 0.3, a line sits 30% of the way from the thinnest to the thickest. | 0.1 to 30 px | 4 px | `Show the reticulations` is **On**, `Width from γ` is **On** |
| `Dash length` | | 0.5 to 50 px | 4 px | `Show the reticulations` is **On** |
| `Gap between dashes` | | 0.5 to 50 px | 3 px | `Show the reticulations` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Style` | Major tree: the tree of the major edges, with each minor edge drawn from the donor to the hybrid node. Full network: each hybrid node is a point of its own, where the major edge (from its parent, in the color of its type) and the minor edge (from the donor, in a lighter shade) arrive as curves, and one branch goes on from there. | Major tree<br>Full network | Major tree | `Show the reticulations` is **On** |
| `Arrows` | Arrow heads toward the hybrid (the receiver), toward the donor, at both ends or none. | None<br>To hybrid<br>To donor<br>Both | To hybrid | `Show the reticulations` is **On** |
| `Arrow head size` | Length of each arrow head. It is always a little wider than the line. | 1 to 80 px | 8 px | `Show the reticulations` is **On**, `Arrows` is not **None** |
| `Line` | | Straight<br>Curved | Curved | `Show the reticulations` is **On**, `Style` is **Major tree** |
| `Curvature` | How far the curve bends toward the root. | 5 to 100% | 40% | `Show the reticulations` is **On**, `Style` is **Major tree**, `Line` is **Curved** |
| `Where on the branches` | Point of each branch where the lines start and end.[1] | Centered<br>Lengths<br>Custom | Centered | `Show the reticulations` is **On** |
| `Position along the branch` | 0% is the parent end of each branch, 100% its node. | 0 to 100% | 50% | `Show the reticulations` is **On**, `Where on the branches` is **Custom** |
| `Show a value on each line` | Writes a value of each reticulation next to its line. | On<br>Off | Off | `Show the reticulations` is **On**, the reticulations carry values |
| `Value` | γ, the length or the support of each reticulation, or any [&...] annotation of its two ends, as written in the file. The minor value goes on the minor edge (the reticulation line). | List | γ (inheritance) | `Show the reticulations` is **On**, `Show a value on each line` is **On** |
| `Decimals` | | 0 to 4 | 2 | `Show the reticulations` is **On**, `Show a value on each line` is **On** |
| `Color the values like their line` | The major value takes the color of its reticulation type (#H, #LGT or #R), the minor value a lighter shade of it. | On<br>Off | On | `Show the reticulations` is **On**, `Show a value on each line` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Minor value: text size` | | 1 to 100 px | 9 px | `Show the reticulations` is **On**, `Show a value on each line` is **On** |
| `Minor value: color` | | Color | ■ #6A6A6A | `Show the reticulations` is **On**, `Show a value on each line` is **On**, `Color the values like their line` is **Off** |
| `Minor value: horizontal offset` | Moves the minor value sideways. | −500 to 500 px | 4 px | `Show the reticulations` is **On**, `Show a value on each line` is **On** |
| `Minor value: vertical offset` | Moves the minor value up (positive) or down (negative). | −500 to 500 px | 3 px | `Show the reticulations` is **On**, `Show a value on each line` is **On** |
| `Also show the γ of the major edge` | Writes the γ of the main parent too: under the hybrid node's branch, or just before the hybrid node in a full network. | On<br>Off | Off | `Show the reticulations` is **On**, `Show a value on each line` is **On**, `Value` is **γ (inheritance)** |
| `Major value: text size` | | 1 to 100 px | 9 px | `Show the reticulations` is **On**, `Show a value on each line` is **On**, `Also show the γ of the major edge` is **On** |
| `Major value: color` | | Color | ■ #6A6A6A | `Show the reticulations` is **On**, `Show a value on each line` is **On**, `Also show the γ of the major edge` is **On**, `Color the values like their line` is **Off** |
| `Major value: horizontal offset` | Moves the major value sideways. | −500 to 500 px | 0 px | `Show the reticulations` is **On**, `Show a value on each line` is **On**, `Also show the γ of the major edge` is **On** |
| `Major value: vertical offset` | Moves the major value up (positive) or down (negative). | −500 to 500 px | 0 px | `Show the reticulations` is **On**, `Show a value on each line` is **On**, `Also show the γ of the major edge` is **On** |

[1] Centered: the middle of each branch. Lengths: where the network places each end, from the branch lengths of the file (without lengths, the middle). Custom: the point chosen with `Position along the branch`.

## Scale bar

The bar that gives the scale of the branch lengths, for trees without a time axis.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show scale bar` [1] | A bar showing a length in branch units. | On<br>Off | On | |
| `Length` | Automatic picks a round length, custom uses your own. | Automatic<br>Custom | Automatic | `Show scale bar` is **On** |
| `Bar length` | | $10^{-9}$ to $10^{9}$ | 0.1 | `Show scale bar` is **On**, `Length` is **Custom** |
| `Unit` | Written after the number and centered with it on the bar, for example Ma, subs./site or any text. | Text | | `Show scale bar` is **On** |
| `Decimals` | | Auto, 0, 1, 2, 3, 4 | Automatic | `Show scale bar` is **On** |
| `Vertical position` | | Top<br>Bottom | Bottom | `Show scale bar` is **On**, the bar is locked on the canvas |
| `Horizontal position` | | Left<br>Center<br>Right | Left | `Show scale bar` is **On**, the bar is locked on the canvas |
| `Move sideways` | | −400 to 400 px | 0 px | `Show scale bar` is **On**, the bar is locked on the canvas |
| `Move up or down` | | −400 to 400 px | 0 px | `Show scale bar` is **On**, the bar is locked on the canvas |
| `Line width` | | 0.1 to 20 px | 1.2 px | `Show scale bar` is **On** |
| `End tick height` | Height of the small marks at both ends. | 0 to 20 px | 4 px | `Show scale bar` is **On** |
| `Text size` | | 1 to 100 px | 10 px | `Show scale bar` is **On** |
| `Text distance from bar` | | −40 to 40 px | 4 px | `Show scale bar` is **On** |
| `Color` | | Color | ■ #333333 | `Show scale bar` is **On** |

[1] Measured in branch length units.

## Time axis

An axis of ages for dated trees, with the geologic time scale, guide lines and shaded intervals.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show time axis (dated trees)` [1] | Adds an axis of ages under the tree. | On<br>Off | Off | `Use branch lengths` turned **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Position` | Below or above the tree, with its ticks and ages outward. On vertical trees, Bottom is on the right and Top on the left. With the geologic time scale, the axis goes with its boxes instead. | Bottom<br>Top | Bottom | `Show time axis (dated trees)` is **On**, rectangular trees, no geologic time scale |

[1] Needs branch lengths in time units.

## › Ages

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Age of the youngest tip` [1] | Age where the axis starts. The ages written in labels count from it too, as the axis does. This includes node and branch age annotations (height, age, and their medians, means and intervals), the ages of tree sets and the ages of the tips. Only the numbers shown change, the tree keeps its own heights. | $-10^6$ to $10^6$ | 0 | `Show time axis (dated trees)` is **On** |
| `Time units per branch length unit` [2] | Factor that turns branch lengths into ages. | $10^{-9}$ to $10^9$ | 1 | `Show time axis (dated trees)` is **On** |
| `Axis title` | | Text | Ma | `Show time axis (dated trees)` is **On** |
| `Show units next to the youngest value` | Writes the units after the first number of the axis. | On<br>Off | Off | `Show time axis (dated trees)` is **On** |
| `Units` | | Text | Ma | `Show time axis (dated trees)` is **On**, `Show units next to the youngest value` is **On** |

[1] Change it when the youngest tip is not from the present, as in a fossil-only tree. All tips then count as extinct.
[2] Multiplies the branch lengths to get ages. Leave it at 1 when the lengths are already in the units of the axis.

## › Ticks

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Place ticks` | How the ages on the axis are chosen. | Automatic<br>Every…<br>Geologic<br>Custom | Automatic | `Show time axis (dated trees)` is **On** |
| `Ages` | Your own list of ages to mark, separated by commas. To show a name instead of the number, add it after an equals sign, for example 66=K–Pg. | Text | | `Place ticks` is **Custom**, `Show time axis (dated trees)` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Interval` | Distance between ticks, in the units of the axis. | $10^{-9}$ to $10^{9}$ | 10 | `Show time axis (dated trees)` is **On**, `Place ticks` is **Every...** |
| `Boundaries of` | Geologic units whose boundaries become ticks. | Era<br>Period<br>Epoch<br>Age | Period | `Show time axis (dated trees)` is **On**, `Place ticks` is **Geologic** |
| `Minor ticks between labels` | Small unlabeled ticks between the numbered ones. | 0 to 9 | 1 | `Show time axis (dated trees)` is **On**, `Place ticks` is not **Custom** |
| `Decimals` | | Auto, 0, 1, 2, 3 | Automatic | `Show time axis (dated trees)` is **On** |

## › Axis look

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Line width` | | 1 to 6 px | 1 px | `Show time axis (dated trees)` is **On** |
| `Tick length` | | 0 to 20 px | 5 px | `Show time axis (dated trees)` is **On** |
| `Show the ages on the circles` | Writes the ages on the rings of circular trees. | On<br>Off | On | **Circular**, **Fan** or **Unrooted** layout, `Show time axis (dated trees)` is **On** |
| `Number size` | | 1 to 100 px | 10 px | **Rectangular** layout, **Circular** or **Fan** layout, `Show time axis (dated trees)` is **On**, `Show the ages on the circles` is **On** |
| `Space between numbers and axis` | | 0 to 30 px | 3 px | `Show time axis (dated trees)` is **On** |
| `Title size` | | 1 to 100 px | 11 px | `Show time axis (dated trees)` is **On** |
| `Distance from the tree` | Space between the tree and the axis. | 0 to 60 px | 8 px | `Show time axis (dated trees)` is **On** |
| `Axis color` | | Color | ■ #333333 | `Show time axis (dated trees)` is **On** |
| `Guide lines at each label` | Lines across the tree at every labeled age. | On<br>Off | Off | `Show time axis (dated trees)` is **On** |
| `Guide line color` | | Color | ■ #9A9A9A | `Show time axis (dated trees)` is **On**, `Guide lines at each label` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Guide line style` | | Solid<br>Dashed<br>Dotted | Dashed | `Show time axis (dated trees)` is **On**, `Guide lines at each label` is **On** |
| `Guide line width` | | 0.05 to 20 px | 0.6 px | `Show time axis (dated trees)` is **On**, `Guide lines at each label` is **On** |
| `Guide dash length` | | 1 to 20 px | 3 px | `Show time axis (dated trees)` is **On**, `Guide lines at each label` is **On**, `Guide line style` is **Dashed** |
| `Guide gap between dashes or dots` | | 1 to 20 px | 3 px | `Show time axis (dated trees)` is **On**, `Guide lines at each label` is **On**, `Guide line style` is not **Solid** |

## › Geologic time scale

Boundaries follow the GSA Geologic Time Scale v. 6.0.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show geologic boxes with official colors` | Adds the geologic time scale under the axis, with the colors of the International Commission on Stratigraphy. | On<br>Off | Off | `Show time axis (dated trees)` is **On** |
| `Position` | Below or above the tree. The boxes take their axis (line, ticks, ages and title) with them, and a second axis can go on the other side. | Bottom<br>Top | Bottom | The geologic time scale is shown, rectangular trees |
| `Ticks and ages with the boxes` | Off leaves the colored boxes alone, without the line, ticks, ages and title of the axis. | On<br>Off | On | The geologic time scale is shown |
| `Axis title with the boxes` | Off takes the title of the axis away while the boxes are shown. | On<br>Off | On | The geologic time scale is shown, `Ticks and ages with the boxes` is **On** |
| `Eras`<br>`Periods`<br>`Epochs`<br>`Ages (stages)` | Rows of the time scale to show. | On<br>Off | Off | `Show time axis (dated trees)` is **On**, `Show geologic boxes with official colors` is **On** |
| `Row height` | | 4 to 40 px | 14 px | `Show time axis (dated trees)` is **On**, `Show geologic boxes with official colors` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show names that fit` | Writes the name of each unit when there is room. | On<br>Off | On | `Show time axis (dated trees)` is **On**, `Show geologic boxes with official colors` is **On** |
| `Name size` | | 1 to 60 px | 8 px | `Show time axis (dated trees)` is **On**, `Show geologic boxes with official colors` is **On**, `Show names that fit` is **On** |
| `Box border width` | | 0 to 10 px | 0.5 px | `Show time axis (dated trees)` is **On**, `Show geologic boxes with official colors` is **On** |
| `Box border color` | | Color | ■ #333333 | `Show time axis (dated trees)` is **On**, `Show geologic boxes with official colors` is **On**, `Box border width` is **On** |
| `Tree background` [1] | Shades the area behind the tree by geologic interval, or marks intervals of your own. | None<br>Colors<br>Stripes<br>Custom | None | `Show time axis (dated trees)` is **On** |
| `Background intervals` | Intervals used to shade the background. | Era<br>Period<br>Epoch<br>Age<br>Ticks | Period | `Show time axis (dated trees)` is **On**, `Tree background` is **Colors** or **Stripes** |
| `Intervals` | Intervals of your own, each written as two ages joined by a dash and separated by commas, for example `66-56, 201.4-145`. To give one its own color, add it after a colon: `66-56:#F2A65A`. The ticks and ages of the axis stay as they are set, so a period can be marked while the axis shows another scale. | Text | | `Show time axis (dated trees)` is **On**, `Tree background` is **Custom** |
| `Draw as` | A band of color over each interval, lines at its two limits, or both. | Band<br>Lines<br>Both | Band | `Show time axis (dated trees)` is **On**, `Tree background` is **Custom** |
| `Band color` | The color of the intervals without one of their own. | Color | #8C8C8C #8C8C8C | `Show time axis (dated trees)` is **On**, `Tree background` is **Custom**, `Draw as` is not **Lines** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Background opacity` | | 0.05 to 1 | 0.25 | `Show time axis (dated trees)` is **On**, `Tree background` is not **None** |
| `Line color` | The lines at the limits of the custom intervals. | Color | ■ #555555 | `Show time axis (dated trees)` is **On**, `Tree background` is **Custom**, `Draw as` is not **Band** |
| `Line width` | | 0.05 to 20 px | 1 px | `Show time axis (dated trees)` is **On**, `Tree background` is **Custom**, `Draw as` is not **Band** |
| `Line style` | | Solid<br>Dashed<br>Dotted | Dashed | `Show time axis (dated trees)` is **On**, `Tree background` is **Custom**, `Draw as` is not **Band** |
| `Extend the background and guide lines down to the axis` | The shading and guides reach the axis instead of stopping at the tree. | On<br>Off | Off | **Rectangular** layout, `Show time axis (dated trees)` is **On**, `Tree background` is not **None**, `Guide lines at each label` is **On** |

[1] **Colors** uses the official colors of the intervals, **Stripes** alternates light gray bands, and **Custom** marks intervals of your own. **Custom** and the background intervals are not shown together.

### › Second axis

A plain second axis with the same ages but its own ticks, for example every 10 above an axis every 50. It can go on either side of the tree and starts on the side opposite the axis already there. It stacks beyond whatever is already on its side. The grid follows the main axis.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show a second axis` | | On<br>Off | Off | `Show time axis (dated trees)` is **On**, rectangular trees |
| `Position` | | Bottom<br>Top | Top | `Show a second axis` is **On** |
| `Every (0 for automatic)` | The interval between its ages. | 0 and up | 0 | `Show a second axis` is **On** |
| `Minor ticks between` | | 0 to 9 | 0 | `Show a second axis` is **On** |
| `Decimals (-1 for automatic)` | | −1 to 6 | −1 | `Show a second axis` is **On** |
| `Title` | | Text | | `Show a second axis` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Distance from the previous element` | Space between the second axis and what is already on its side. | 0 to 200 px | 8 px | `Show a second axis` is **On** |

## Ancestral reconstructions

Ancestral states from a SIMMAP file, shown as pie charts on the nodes and colored branches.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `States` | Colors of the states found in the SIMMAP maps. | One color per state found in the maps | | A SIMMAP file is attached |
| `Show pies on the nodes` | Pie charts with the probability of each state. | On<br>Off | On | |
| `Show on` [1] | Which nodes get pies. | All nodes<br>Internal<br>Tips | Internal | `Show pies on the nodes` is **On** |
| `Place` | Pies on the node, or in the middle of the branch that leads to it. | On the node<br>Middle of the branch | On the node | `Show pies on the nodes` is **On** |
| `Group the small slices` | Joins the least probable states into one slice. | On<br>Off | Off | |
| `Group the slices below` | States under this probability are grouped. | 1 to 50% | 10% | `Group the small slices` is **On** |
| `Color of the group` [2] | | Color | #D9D9D9 | `Group the small slices` is **On** |

[1] Size, border and opacity of the pies are set in Nodes. A pie replaces any dot or support circle on its node.
[2] The slices below the threshold are joined into one slice of this color, in the pies and on the branches, the legend calls it "Other".

### › Branches

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Color the branches` | Paints the states along each branch, from the SIMMAP maps. | No<br>Maximum a posteriori<br>Most probable states<br>Probability gradient | No | |
| `Ignore changes shorter than` | Short changes of state are merged into their neighbors. | 1 to 50% | 10% | `Color the branches` is **Maximum a posteriori** |
| `Dominance threshold` [1] | Probability a state needs to be drawn alone. | 1 to 100% | 50% | `Color the branches` is **Most probable states** |
| `Exclusion threshold` [2] | Probability a state needs to appear when none dominates. | 0 to 100% | 25% | `Color the branches` is **Most probable states** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Stripe length` | Length of the stripes when several states share a stretch. | 1 to 20 px | 3 px | `Color the branches` is **Most probable states** |
| `Show a legend` | | On<br>Off | On | |

[1] Each branch is split into short stretches, and each stretch is drawn with the states whose posterior probability there reaches this threshold. With 50%, a state present in at least half of the maps is drawn alone.
[2] Used only where no state reaches the dominance threshold. There, every state at or above this threshold is drawn, alternating as stripes, and the less probable ones are left out. When no state reaches it either, the most probable state is drawn.

## Attachments

Tables (CSV or TSV) and other files that add data to the tree. Rows are matched to tips and internal nodes by name. When another tree is opened while files are attached, tree2go says how many tips of the new tree each file matches, and asks whether to keep them or remove them all. Projects can be saved with the attached files (tables, SIMMAP, MCMC log, alignments, pie charts and images), so that anyone who opens them sees the same figure, or without them, and the window shows the size of each.

Attached tables are never used automatically. Every plot and every style drawn from a table starts on None until you choose it, so a table that matches a tree by chance (a tip, or support values read as names of internal nodes) changes nothing.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Add file` | Adds data tables to the tree, as CSV, TSV or text files whose first column holds the names. It also takes FASTA alignments, which are listed with the other files and shown from Side plots, Alignment. | File | | |
| `Table list` | The attached tables, with how many names of each one match the tree and a button to remove it. | List | | A table is added |
| `MCMC log` | Posterior of each sampled tree, needed for the MAP tree. Use the BEAST2 `.log` or the MrBayes `.p` file; for trees from several runs, the `.p` file of each run, which can be chosen together. | File | | A trees file is open |

## Side plots

Bars, categories and a heatmap drawn next to the tips, from the attached tables. Their options appear once a table that matches the tree is chosen.

### › Layers

The side plots are layers next to the tips. The first in the list is the nearest to the tree, and each one starts where the one before it ends. Drag a layer or use its arrows to move it, and click its eye to hide it. A hidden layer keeps its table and settings but is not drawn and takes no room, and its options are grayed out until it is shown again. Layers not in use are listed grayed out as *not in use*, and layers that the layout of the tree does not draw (the alignment and the heatmap on circular and fan trees) as *not visible*. With the clade names first, their bars go right after the tip names. On circular and fan trees the clade name bars always go around the outside, whatever their place in the list.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Layers` | The order of the side plots from the tips outward, and which ones are shown. | Alignment<br>Bars<br>Categories<br>Heatmap<br>Clade names | Alignment > Bars > Categories > Heatmap > Clade names | Always |

## › Bars

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Table (bars)` | Table with the values to draw as bars. | A table that matches the tree | None | A table is attached |
| `Columns (bars)` | Numeric columns drawn as bars, each with its color. | List | | A table is chosen for the bars |
| `Width of the bar area` | Room for the longest bar. | 20 to 600 px | 150 px | A table is chosen for the bars |
| `Distance from the previous element` | Space between the bars and the layer before it in `Layers`. Changing it moves the bars and every layer after it together, as one block, and making it smaller brings them back the same way. | 0 to 200 px | 12 px | A table is chosen for the bars |
| `Bar thickness` | Share of the row each bar takes. | 10 to 100% | 70% | A table is chosen for the bars |
| `Bar opacity` | | 5 to 100% | 100% | A table is chosen for the bars |
| `Color the bars by named clade` | Each bar takes the color of its clade name bar. | On<br>Off | Off | A table is chosen for the bars |
| `Show as proportions` | Stacks the columns as parts of a whole. | On<br>Off | Off | A table is chosen for the bars |
| `Show them as` | Writes the parts as proportions or percentages. | Proportion<br>Percentage | Proportion | A table is chosen for the bars, `Show as proportions` is **On** |
| `Show the values next to the bars` | | On<br>Off | Off | A table is chosen for the bars, `Show as proportions` is **On** |
| `Value size` | | 1 to 100 px | 9 px | A table is chosen for the bars, `Show the values next to the bars` is **On**, `Show as proportions` is **On** |
| `Upper limit of the bars` | Value at the end of the bar area, empty uses the highest value. | 0 to $10^{12}$ | | A table is chosen for the bars |
| `Show a legend` | | On<br>Off | On | A table is chosen for the bars |

## › Bar scale

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show a scale under the bars` | Adds an axis with the values of the bars. | On<br>Off | On | A table is chosen for the bars, **Rectangular** layout |
| `Distance from the bars` | | 0 to 60 px | 4 px | A table is chosen for the bars, **Rectangular** layout, `Show a scale under the bars` is **On** |
| `Line width` | | 0.2 to 4 px | 0.8 px | A table is chosen for the bars, **Rectangular** layout, `Show a scale under the bars` is **On** |
| `Number distance from the line` | | 0 to 30 px | 2 px | A table is chosen for the bars, rectangular layout, `Show a scale under the bars` is **On** |
| `Tick length` | | 0 to 12 px | 4 px | A table is chosen for the bars, **Rectangular** layout, `Show a scale under the bars` is **On** |
| `Number size` | | 1 to 100 px | 9 px | A table is chosen for the bars, **Rectangular** layout, `Show a scale under the bars` is **On** |
| `Color` | | Color | ■ #333333 | A table is chosen for the bars, rectangular layout, `Show a scale under the bars` is **On** |

## › Categories

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Table (categories)` | Table with the categories to draw as shapes. | A table that matches the tree | None | A table is attached |
| `Columns (categories)` | Columns drawn as shapes, filled where present. | List | | A table is chosen for the categories |
| `Shape` | | Squares<br>Circles | Squares | A table is chosen for the categories |
| `Size` | | 4 to 40 px | 10 px | A table is chosen for the categories |
| `Space between columns` | | 0 to 40 px | 3 px | A table is chosen for the categories |
| `Distance from the previous element` | Space between the categories and the layer before it in `Layers`. Changing it moves the categories and every layer after it together, as one block, and making it smaller brings them back the same way. | 0 to 200 px | 12 px | A table is chosen for the categories |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Absent shapes` | How absent values are drawn. | Filled<br>Outline only | Filled | A table is chosen for the categories |
| `Absent color` | | Color | #E2E2E2 | A table is chosen for the categories |
| `Shape border` | | Solid<br>Dashed<br>Dotted<br>None | None | A table is chosen for the categories |
| `Border color` | | Color | #1F1F1F | A table is chosen for the categories, `Shape border` is not **None** |
| `Border width` | | 0.05 to 50 px | 1 px | A table is chosen for the categories, `Shape border` is not **None** |
| `Number of dashes or dots` | | 1 to 60 | 12 | A table is chosen for the categories, `Shape border` is **Dashed** or **Dotted** |
| `Turn circular and fan trees inside out` | Puts the tree outside and the categories inside the circle. | On<br>Off | Off | A table is chosen for the categories, **Circular** or **Fan** layout |
| `Show the column titles` | | On<br>Off | On | A table is chosen for the categories, **Rectangular** layout |
| `Title angle` | | 45°<br>Vertical | 45° | A table is chosen for the categories, **Rectangular** layout, `Show the column titles` is **On**, `Orientation` is not **Vertical** |
| `Title typeface` | | Same as the labels | | A table is chosen for the categories, **Rectangular** layout, `Show the column titles` is **On** |
| `Titles in bold` | | On<br>Off | Off | A table is chosen for the categories, **Rectangular** layout, `Show the column titles` is **On** |
| `Titles in italic` | | On<br>Off | Off | A table is chosen for the categories, **Rectangular** layout, `Show the column titles` is **On** |
| `Title size` | | 1 to 100 px | 10 px | A table is chosen for the categories, **Rectangular** layout, `Show the column titles` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Title distance` | | 0 to 40 px | 6 px | A table is chosen for the categories, **Rectangular** layout, `Show the column titles` is **On** |
| `Title color` | | Color | ■ #333333 | A table is chosen for the categories, **Rectangular** layout, `Show the column titles` is **On** |

## › Heatmap

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Table (heatmap)` | Table with the values of the heatmap. | A table that matches the tree | None | A table is attached |
| `Columns (heatmap)` | Numeric columns, one column of colored cells each. | List | | A table is chosen for the heatmap |
| `Color scale` | One color scale per column, or one for all columns. | Each column<br>All columns | Each column | A table is chosen for the heatmap |
| `Low values` | | Color | ■ #2C7BB6 | A table is chosen for the heatmap |
| `Use a middle color` | Adds a third color for values in the middle of the range. | On<br>Off | On | A table is chosen for the heatmap |
| `Middle values` | | Color | ■ #FFFFBF | A table is chosen for the heatmap, `Use a middle color` is **On** |
| `High values` | | Color | ■ #D7191C | A table is chosen for the heatmap |
| `Leave missing values empty` | Missing values get their own color. | On<br>Off | Off | A table is chosen for the heatmap |
| `Missing values` | | Color | ■ #E5E5E5 | A table is chosen for the heatmap, `Leave missing values empty` is **Off** |
| `Cell width` | | 2 to 60 px | 14 px | A table is chosen for the heatmap |
| `Cell height` | Share of the row each cell takes. | 0.1 to 1 × row | 0.9 × row | A table is chosen for the heatmap |
| `Space between columns` | | 0 to 20 px | 1 px | A table is chosen for the heatmap |
| `Distance from the previous element` | Space between the heatmap and the layer before it in `Layers`. Changing it moves the heatmap and every layer after it together, as one block, and making it smaller brings them back the same way. | 0 to 100 px | 10 px | A table is chosen for the heatmap |
| `Column titles` | | On<br>Off | On | A table is chosen for the heatmap |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Title angle` | | 0 to 90° | 60° | A table is chosen for the heatmap, `Column titles` is **On**, `Orientation` is not **Vertical** |
| `Title size` | | 1 to 100 px | 9 px | A table is chosen for the heatmap, `Column titles` is **On** |
| `Title color` | | Color | ■ #333333 | A table is chosen for the heatmap, `Column titles` is **On** |
| `Show legend` | | On<br>Off | On | A table is chosen for the heatmap |

## › Alignment

A sequence alignment (FASTA) next to the tips, matched by name. The cells are square, as tall as the rows, so the width decides how many sites fit. When more sites are asked than fit, the rows become a barcode where each column is compared with its most common residue. Protein alignments are detected automatically. When fewer than 90% of the letters (gaps and N aside) are A, C, G, T or U, the file is read as protein. Their residues are colored by chemistry, in eight groups like the Clustal scheme, each with its own color: hydrophobic, positive, negative, polar, cysteine, glycine, proline and aromatic. Ambiguous letters (X, B, Z…) take the color for other or ambiguous ones.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Alignment` | The alignment shown next to the tips, among the ones loaded in Attachments. A loaded alignment is not shown until it is chosen here. | None<br>The alignments loaded | None | An alignment is loaded |
| `From site` | | 1 to the length | 1 bp | `Alignment` is not **None** |
| `To site` | | 1 to the length | Last bp | `Alignment` is not **None** |
| `Width` | Space for the alignment, which sets how many sites fit as squares. | 20 to 2000 px | 300 px | `Alignment` is not **None** |
| `Cell size (% of the row)` | | 20 to 100% | 100% | `Alignment` is not **None** |
| `Distance from the previous element` | Space between the alignment and the layer before it in `Layers`. Changing it moves the alignment and every layer after it together, as one block, and making it smaller brings them back the same way. | 0 to 200 px | 10 px | `Alignment` is not **None** |
| `Site numbers` | | On<br>Off | On | `Alignment` is not **None** |
| `Place` | Site numbers above or below the alignment. | Above<br>Below | Above | `Alignment` is not **None**, `Site numbers` is **On** |
| `Every (sites, 0 for automatic)` | How many sites between two numbers. | 0 to 100000 | 0 (automatic) | `Alignment` is not **None**, `Site numbers` is **On** |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Text size` | | 4 to 40 px | 8 px | `Alignment` is not **None**, `Site numbers` is **On** |
| `Space between numbers and ticks` | | 0 to 40 px | 2 px | `Alignment` is not **None**, `Site numbers` is **On** |
| `Tick length` | | 0 to 30 px | 4 px | `Alignment` is not **None**, `Site numbers` is **On** |
| `Color of numbers and ticks` | | Color | #555555 | `Alignment` is not **None**, `Site numbers` is **On** |
| `Letters in the cells (when large enough)` | | On<br>Off | On | `Alignment` is not **None** |
| `Adenine (A)` | | Color | #5CB85C | `Alignment` is not **None**, a DNA alignment |
| `Cytosine (C)` | | Color | #4A86C5 | `Alignment` is not **None**, a DNA alignment |
| `Guanine (G)` | | Color | #F0AD4E | `Alignment` is not **None**, a DNA alignment |
| `Thymine (T) or uracil (U)` | | Color | #D9534F | `Alignment` is not **None**, a DNA alignment |
| `Hydrophobic (A, I, L, M, F, W, V)` | | Color | #80A0F0 | `Alignment` is not **None**, a protein alignment |
| `Positive (K, R)` | | Color | #F01505 | `Alignment` is not **None**, a protein alignment |
| `Negative (D, E)` | | Color | #C048C0 | `Alignment` is not **None**, a protein alignment |
| `Polar (N, Q, S, T)` | | Color | #15C015 | `Alignment` is not **None**, a protein alignment |
| `Cysteine (C)` | | Color | #F08080 | `Alignment` is not **None**, a protein alignment |
| `Glycine (G)` | | Color | #F09048 | `Alignment` is not **None**, a protein alignment |
| `Proline (P)` | | Color | #C0C000 | `Alignment` is not **None**, a protein alignment |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Aromatic (H, Y)` | | Color | #15A4A4 | `Alignment` is not **None**, a protein alignment |
| `Other or ambiguous` | | Color | #B0B0B0 | `Alignment` is not **None** |
| `Gaps and missing data` | | Color | #F2F2F2 | `Alignment` is not **None** |
| `Barcode: as the most common` | | Color | #C8C8C8 | `Alignment` is not **None**, more sites than fit |
| `Barcode: different` | | Color | #1F1F1F | `Alignment` is not **None**, more sites than fit |

## Legends

Placement and look of every legend or scale bar. Hover an element on the canvas and click its lock to move it by hand. `Right-click` it to rewrite its texts.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show legends` | Shows or hides every legend of the figure. It shows a tick when all are shown, a dash when only some are, and is empty when none are. A click on a dash shows them all. | On<br>Off | On | |
| List of legends | Every legend the figure has, as the layers of the side plots, each with an eye to show or hide it. A legend hidden from its `Right-click` menu ( `Hide this legend` ) comes back from here. Hidden legends keep their place and look, and are left out of exports. | On<br>Off | On | The figure has legends |

### › Place

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Vertical position` | Where the legends sit when they are not placed by hand. | Top<br>Bottom | Bottom | |
| `Horizontal position` | Where the legends sit when they are not placed by hand. | Left<br>Center<br>Right | Left | |
| `Move sideways` | | −400 to 400 px | 0 px | |
| `Move up or down` | | −400 to 400 px | 0 px | |

### › Look

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Text size` | | 1 to 100 px | 10 px | |
| `Title size` | | 1 to 100 px | 10 px | |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Bold titles` | | On<br>Off | On | |
| `Symbol size` | Size of the symbols relative to the text. | 0.5 to 3× | 1× | |
| `Space between symbol and text` | | 0 to 40 px | 8 px | |
| `Space between rows` | | 0 to 30 px | 4 px | |
| `Space between legends` | Space between legends stacked together. | 0 to 60 px | 14 px | |
| `Text color` | | Color | ■ #333333 | |

## Split into pages

Split a long rectangular tree into pages of the same width, each with a small map of the whole tree.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Pages` | The pages made so far, from top to bottom. | List | | The tree has pages |
| `Split the rest into pages` | Makes pages for every row that has none. | | | |
| `Split the tree into … pages` | Splits the whole tree into this number of pages. | Number | 2 | |
| `Save pages as PDF` | One PDF with a page per section. | | | The tree has pages |
| `PNG (.zip)`<br>`SVG (.zip)` | One image per page, in a `.zip`. | | | The tree has pages |

### › Look

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Page box on the canvas` | Color of the page frames on screen (for visualization only, they are not exported). | Color | ■ #27408B | **Rectangular** layout, the tree has pages |
| `Repeat the time axis under each page` | Each page gets its own copy of the axis. | On<br>Off | On | **Rectangular** layout, the tree has pages, `Show time axis (dated trees)` is **On** |
| `Include the title, legends and scale bar` | The first page gets the title and the last one the legends and scale bar. | On<br>Off | Off | **Rectangular** layout, the tree has pages |
| `Extra width on the left` | Widens every page to the left, the same for all. | −200 to 600 px | 0 px | **Rectangular** layout, the tree has pages |
| `Extra width on the right` | Widens every page to the right, the same for all. | −200 to 600 px | 0 px | **Rectangular** layout, the tree has pages |

### › Tree map

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show a map of the whole tree on each page` | A small tree with the rows of the page highlighted. | On<br>Off | On | **Rectangular** layout, the tree has pages |
| `Map width` | | 20 to 300 px | 60 px | **Rectangular** layout, the tree has pages, `Show a map of the whole tree on each page` is **On** |
| `Map height` | | 20 to 600 px | 120 px | **Rectangular** layout, the tree has pages, `Show a map of the whole tree on each page` is **On** |
| `Space to the tree` | Space between the map and the tree. | 0 to 200 px | 24 px | **Rectangular** layout, the tree has pages, `Show a map of the whole tree on each page` is **On** |
| `Line width` | | 0.1 to 10 px | 0.8 px | **Rectangular** layout, the tree has pages, `Show a map of the whole tree on each page` is **On** |
| `Tree color` | | Color | #BDBDBD | **Rectangular** layout, the tree has pages, `Show a map of the whole tree on each page` is **On** |
| `Color of the rows on the page` | | Color | #C8553D | **Rectangular** layout, the tree has pages, `Show a map of the whole tree on each page` is **On** |
| `Mark the rows on the page with` | How the rows of the page stand out on the map. | Branch color<br>Color and background | `Color of the rows on the page` | **Rectangular** layout, the tree has pages, `Show a map of the whole tree on each page` is **On** |

## Geography

Maps for the figure, from a file of your own or downloaded, with the records of a table drawn on them. Each map is an image that moves, aligns and shows guides like any other, numbered, and drawn behind the tree. Its `Right-click` menu sets its name, its lines, fill, the area shown (the whole map, only around the points with a margin, or a custom view), its scale as 1:X, a frame and coordinates. Its proportions can be unlocked (`Position and size`, `Proportions`). The frame then takes any shape, and the map shows more or less of its area to fill it, centered and never stretched. With the map selected, `Ctrl` + drag moves its view inside its frame and `Ctrl` + scroll zooms in and out, as in a GIS, and the area becomes custom. Each map has its own points, projection, styles and lines. The options below `Map` belong to the map chosen there, and appear

once a map is on the figure. Points on GPlates coastline maps are automatically reconstructed to the map's age and plate model. The map and its points must use the same coordinate system, either both in degrees (WGS84), projected together, or both in the same planar system (meters), drawn as they are. When they may not match (latitudes beyond ±90°, degrees against meters, or most points outside the map), a window says so, with what the `.prj` file of the map says.

## › Add a map

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Map from a shapefile…` | Reads a `.shp` file, or a `.zip` with its `.shp`, `.dbf` and `.prj`. Very detailed maps are simplified as they load. | | | Always |
| `Load from Natural Earth` | Downloads one layer (land, countries or coastlines) at the resolution chosen, as a map. Once loaded, the map is kept in the app, and in projects saved with their attached files. | 1:110 million<br>1:50 million<br>1:10 million<br>Land<br>Countries<br>Coastlines | 1:110 million, land | Internet connection available |
| `Load the Coastlines of the past (GPlates)` | Downloads the coastlines of an age and a plate model as a map (0 to 1000 Ma). Its points are reconstructed to the same age and model. | Scotese 2016 (PALEOMAP)<br>Müller et al. (2019)<br>Merdith et al. (2021) | 50 Ma, Scotese 2016 (PALEOMAP) | Internet connection available |
| `Maps loaded` | Every map loaded, each in its own box, with its number of shapes and its units (degrees or planar units). A map deleted from the figure stays loaded. `Show it on the figure` brings it back, and `Add it again as a new map` adds another copy with settings of its own. `Remove` unloads it. | | | A map is loaded |

## › Map

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Map` | The map whose options are shown below. | None<br>The maps on the figure | The last one added | Always |
| `Projection (maps in degrees)` | How a map in degrees, and its points, are drawn on the plane. Maps in a planar system are drawn as they are. | Equal Earth (EPSG:8857)<br>Equirectangular (EPSG:4326)<br>Mercator (EPSG:3857)<br>Mollweide (ESRI:54009) | Equal Earth (EPSG:8857) | A map is on the figure |

## › Coordinates of a map

From the `Right-click` menu of a map you can add ticks with their coordinates on the chosen edges, outside its frame (shown with them). In degrees, each meridian and parallel is placed where it crosses the frame, so they follow any projection and a custom view. Where they would overlap, as at the top of Mollweide, only those with room are written. Planar maps get the coordinates in their own units.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Edges` | Where the coordinates go. | Bottom<br>Left<br>Top<br>Right | None | Always |
| `Every (0 for automatic)` | Degrees (or units) between two ticks. | 0 and up | 0 | An edge is chosen |
| `Format` | 45.50°W, −45.50° (the sign gives the side, without letters), or 45°30′00″W, always written in full. | Decimal degrees (N, S, E, W)<br>Decimal degrees (+ and −)<br>Degrees, minutes, seconds | Decimal degrees | An edge is chosen, a map in degrees |
| `Decimals` | | 0 and up | 0 | An edge is chosen |
| `Tick length` and `Tick width` | | px | 4 px and 0.8 px | An edge is chosen |
| `Space to the numbers` | | px | 2 px | An edge is chosen |
| `Text size` and `Color` | | px<br>Color | 9 px, ■ #333333 | An edge is chosen |

## › Points

One row per record, with the species in the first column, as attached in Attachments. Records at the same place are drawn once. Larger points are drawn behind smaller ones, so they never hide them. With `Info on hover`, a point shows its species, its longitude (x), its latitude (y) and how many records it holds. A click selects a point and `Shift-click` adds or removes points. The `Right-click` menu gives the point, or all the selected ones, a style of their own (color, size, shape, border), which wins over the panel.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Table` | The table of the records. | None<br>The tables attached | None | Always |
| `Longitude (x)` | Found by its name (lon, longitude, x…) unless chosen. | Found by its name<br>The columns of the table | Found by its name | `Table` is chosen |
| `Latitude (y)` | Found by its name (lat, latitude, y…) unless chosen. | Found by its name<br>The columns of the table | Found by its name | `Table` is chosen |
| `Lineages` | All the records, or only those of living or of fossil lineages (as the tips of the tree are found). | All<br>Living<br>Fossils | All | `Table` is chosen |
| `Shape` | | Circle<br>Square<br>Triangle<br>Diamond<br>Star | Circle | `Table` is chosen |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Size` | One size, by how many records a point holds, or by a value. | One size<br>By the number of records<br>By a value | One size | `Table` is chosen |
| `Point size` | With grouping, a grouped point grows with its records, up to `Largest`. | 0.5 to 40 px | 3.5 px | `Table` is chosen, `Size` is **One size** |
| `Value for the size` | A column of the records, a tip annotation of the tree, or a column of another table (by species). | None<br>The columns and annotations found | None | `Table` is chosen, `Size` is **By a value** |
| `Smallest` | | 0.5 to 40 px | 2.5 px | `Table` is chosen, `Size` is not **One size** |
| `Largest` | | 0.5 to 60 px | 12 px | `Table` is chosen, `Size` is not **One size**, or with grouping |
| `Color` | One color, the color of the named clade of the species, the color of the highlight of its clade, a color per category, or a gradient along a value. Species outside any named or highlighted clade, or without a value, are gray. | One color<br>By named clade<br>As the highlight of its clade<br>By a category<br>By a value (gradient) | One color | `Table` is chosen |
| `Point color` | | Color | #C8553D | `Table` is chosen, `Color` is **One color** |
| `Value for the color` | Works like the value for the size but is independent of it, so the color and the size can follow different values. | None<br>The columns and annotations found | None | `Table` is chosen, `Color` is **By a category** or **By a value** |
| `Palette` | | The palettes of the app | Viridis | `Table` is chosen, `Color` is **By a value** |
| `Reverse the colors` | | On<br>Off | Off | `Table` is chosen, `Color` is **By a value** |
| `Opacity` | | 5 to 100% | 85% | `Table` is chosen |
| `Border color` | | Color | #FFFFFF | `Table` is chosen |
| `Border width` | | 0 to 5 px | 0.6 px | `Table` is chosen |
| `Clear the styles of single points` | Every point follows the panel again. | | | `Table` is chosen |

## › Lines to the tips

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Join each tip to its points` | One line from each tip to every point holding a record of its species, on any layout of the tree. The lines follow the map when it moves, pass under the names of the tree and over their map. | On<br>Off | Off | `Table` is chosen |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `From` | | The tip<br>The end of its name | The tip | `Table` is chosen, `Join each tip to its points` is **On** |
| `Line color` | One color, or the color of the point each line goes to. | One color<br>As its point | One color | `Table` is chosen, `Join each tip to its points` is **On** |
| `Color of the lines` | | Color | #8A8F98 | `Table` is chosen, `Line color` is **One color** |
| `Line width` | | 0.1 to 6 px | 0.6 px | `Table` is chosen, `Join each tip to its points` is **On** |
| `Line opacity` | | 5 to 100% | 70% | `Table` is chosen, `Join each tip to its points` is **On** |
| `Line style` | | Solid<br>Dashed<br>Dotted | Solid | `Table` is chosen, `Join each tip to its points` is **On** |

### › Grouping

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Neighborhood radius (map units, 0 for none)` | Records closer than this become one point. It uses the map units, so degrees for maps in degrees, and otherwise the units of the planar system (usually meters). | 0 and up | 0 | `Table` is chosen |
| `Number of records on grouped points` | | On<br>Off | On | `Table` is chosen, `Neighborhood radius` is above 0 |
| `Color of groups with several colors` | A grouped point whose records have different colors takes this one. | Color | #9AA0A6 | `Table` is chosen, `Neighborhood radius` is above 0 |

## Images and visual elements

Pictures (PNG or JPG) placed on the page[1].

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Add image` | Places a PNG or JPG on the page, for example a silhouette. | | | |
| `Add label` | A text you can place anywhere, for example a name inside the tree. It moves, aligns with branches, labels and clades, shows distance guides and takes arrows like an image, and is exported as text. `Right-click` it to write it and choose its size, color, bold, italic and typeface. | | | A tree is loaded |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Add density plot` | A density plot (or histogram) of a value of the tree or of a table, drawn by the app as an image that moves, resizes and aligns like the others. `Right-click` it to choose the values and its look. `Values from` picks the origin: **From tree** (branch lengths and Support values when the nodes have them), **Tree annotations** (such as rates or HPD) or **Attached files** (a column of a table). Its fill can follow the color scale of the branches or of the dots. | | | A tree is loaded |
| `Add LTT plot` | A lineages-through-time plot of a tree with ages, drawn by the app as an image that moves, resizes and aligns like the others. It counts lineages cumulatively (every lineage that has appeared) or alive at each time, at every event or in time bins, and can hold several lines, each for the whole tree or a named clade, with or without fossils and with its own name in the legend. The tree can show faintly behind, and the plot can be stretched freely. Its time axis takes the ages of the tips, or the age of the youngest tip set in Time axis. `Right-click` it to add lines and change its look. | | | A tree is loaded |
| `Draw arrow` | An arrow to point at something. Drag it by its body to move it, or drag one of its ends to anchor it to a node, to the start or the end of the name of a tip, to the start, middle or end of a clade name bar or to its name, or to any of the nine points of an image or of any other visual element (text labels, density and LTT plots, maps). An anchored end follows what it is attached to when it moves. `Right-click` it for its look. | | | A tree is loaded |
| `Image list` | Click a row to select its image. | | | An image is added |
| `Replace` | Swaps the picture, keeping its place and width. | | | An image is added |
| `Remove` | | | | An image is added |

[1] Drag an image to move it, drag a corner to resize it (proportions are kept), or just outside a corner to rotate it. `Right-click` it for position, shape, size, opacity, rotation, mirror, layer and alignment. On rectangular trees, dotted guides show the distance to the nearest branch, tip label or axis, click a number to type an exact distance. Arrow keys move it 1 px (10 px with `Shift`). To align it, select the image, `Shift-click` a branch, label or clade, and `Right-click` the image for more options. Alignment works better with rectangular views.

## Tanglegram

This tree and another one face to face, their tips joined by lines. The second tree is another tab, with its own look and highlights. To get one, open a tree of the set in a tab, load a file in a new tab, or copy this tree to a new tab (`Right-click` its

root, `Copy tree to a new tab`) and change it there. Both trees must have the same tips, or the box says how many are found in one only, and the tanglegram waits until they match.

It needs a rectangular tree that is not turned, with the space to the right of the tree free of side plots, clade name bars, images or other visual elements (these can be moved below or above the tree). The note says what is in the way. With highlights of the same colors on both sides, the two trees read as one.

`Untangle` turns the nodes of the second tree (or of both) one by one, keeping each turn that lowers the number of crossing lines, until none lowers it any more. It writes how many crossings there were and are left.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Show a tanglegram` | | On<br>Off | Off | The conditions above are met |
| `Second tree` | The tab drawn on the right. | The other tabs | The first other tab | A second tab is open |

## › Lines

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `From` | Where the lines start and end, at the end of each name (or each tip) on each side. | The names<br>The tips | The names | `Show a tanglegram` is **On** |
| `Color` | One color, or the color of the highlight (or of the branch) each tip is in, on each side. | One color<br>As the highlights<br>As the branches | One color | `Show a tanglegram` is **On** |
| `Where the colors differ` | When both sides have a color and they differ, use a gradient from one to the other, or one neutral color. A tip with a color on one side only takes that color, and one with none the line color. | From one to the other<br>One neutral color | From one to the other | `Color` is not **One color** |
| `Width`, `Opacity` and `Style` | | px, %<br>Solid, Dashed, Dotted | 0.8 px, 80 %, Solid | `Show a tanglegram` is **On** |
| `Curvature (0 for straight)` | Bends the lines into an S. | 0 to 100 % | 0 | `Show a tanglegram` is **On** |

A click selects a line (it shows in blue), like the branches. `Right-click` a line to give it a color, width or style of its own, and `Back to the look of all` to undo it. The second tree keeps the look of its tab, aligned names with their guides included.

## › Placement

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Distance between the trees` | The room for the lines. | 10 to 2000 px | 140 px | `Show a tanglegram` is **On** |
| `Width (0 as this tree)` | | 0 to 5000 px | 0 | `Show a tanglegram` is **On** |
| `Names of its tips` | Written facing the lines. | On<br>Off | On | `Show a tanglegram` is **On** |

| Option | Description | Values | Default | Appears when |
| --- | --- | --- | --- | --- |
| `Join the highlights of the same clade` | A clade highlighted in both trees, with the very same tips, is joined across the gap between them, in its color when both sides share it, or from one color to the other. A highlight that fades toward the lines joins from its faded edge. | On<br>Off | On | `Show a tanglegram` is **On** |

# Header menus

> The header menus handle the work as a whole, such as opening and saving projects and styles, the page settings, and exporting the figure, the tree or its trees.

## Import and Save

The `Import` ▾ and `Save` ▾ menus of the header open or save a project or a style.

| Item | What it holds |
|---|---|
| `Project` | The trees and everything that goes with them, including tabs, look, tables, alignments, maps, images and, if chosen, every tree of a set. |
| `Style` | Every option of the left panel that can go to another tree, such as the look of the branches, labels, axes and clade name bars. Options tied to one tree (its clades, tables, maps and images) stay out. A style applied to another tree sets those options there, so clade bars named later already take that look. |

### Saving a project

The window shown when a project is saved.

| Option | Description | Default |
|---|---|---|
| `With the attached files` or `Without them` | With them, anyone who opens the project sees the same figure. Without them, the files have to be attached again after opening it. | With the attached files |
| `With the N trees of the set` | Every tree of the set. Opened again, another burn-in, consensus, tree of the set, DensiTree or export of the trees needs no file. Without them, the project keeps the summary tree, with its supports and ages. | On |
| `Without indentation` | Lighter, still plain text. | On |
| `Compressed` | Much lighter, often many times, but not human-readable. A compressed project ends in `.tree2go.json.gz` and opens as any other. | On |

## Page settings

The `Page settings` menu of the header holds the plot title, the background, the palette, the color blind preview and the lightening of heavy elements, then `Copy style`, the only part that folds, and `Reset styles`.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Plot title` | A title over or under the figure. Its other options appear once it has a text. | Text | | |
| `Size` | | 4 to 300 px | 18 px | `Plot title` is set |
| `Vertical position` | | Top<br>Bottom | Top | `Plot title` is set |
| `Horizontal position` | | Left<br>Center<br>Right | Left | `Plot title` is set |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Move sideways` and `Move up or down` | | −400 to 400 px | 0 px | `Plot title` is set |
| `Bold` and `Italic` | | On<br>Off | On, Off | `Plot title` is set |
| `Color` | | Color | ■ #1F1F1F | `Plot title` is set |
| `Background color` | | Color | ▢ #FFFFFF | |
| `Palette` | Colors offered in the menus and for new categories. Colors already in use do not change. | Tree2go<br>Okabe-Ito (color blind safe)<br>Paul Tol bright (color blind safe)<br>Viridis (color blind safe)<br>Inferno (color blind safe)<br>Magma (color blind safe)<br>Plasma (color blind safe)<br>Cividis (color blind safe) | Tree2go | |
| `Color blind preview` | Shows the figure as seen with a color vision deficiency, on screen only. | Normal vision<br>Deuteranopia (red-green)<br>Protanopia (red-green)<br>Tritanopia (blue-yellow)<br>No color (grayscale) | Normal vision | |
| `Render optimization` | Lets you view and edit very large trees (more than 2,000 tips) quickly by drawing only the elements that are in view, or large enough to be seen. Names too small to be read, and names and click areas far from the view, are left out until the view gets to them. Branches are always drawn. Exports always draw everything. Turning it off can lower the performance. | On<br>Off | On | |
| `Very thin lines on screen` | Zoomed far out, lines that would vanish are drawn one pixel wide (on screen only), or keep their true width, as they will look in a small print of the figure. | Always visible<br>As exported | Always visible | |
| `Lighten heavy elements while working` [1] | Shows images, density and LTT plots and maps as light placeholders, to keep the app quick. | On<br>Off | Off | |

[1] Lightened elements stay where they are and can be moved, but not resized. Each shows a gray snapshot of how it last looked, framed by a gray dashed line. They are still there, and exports draw them in full with their current options. One element can also be lightened, or shown again, from its `Right-click` menu (`Lighten this one`, `Show this one`). The box

shows a tick when all are lightened, a dash when only some are, and is empty when none are. A click on a dash lightens them all, and a click on a tick shows them all again.

### › Copy style

What `Paste style` applies, by groups of properties. It copies the look of each kind of element (colors, widths, sizes, places), never their values or data. Unticked groups stay as they are on the elements a style is pasted on, so only the color of the branches can be pasted, for example, or the branches and the nodes but not the labels.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Groups of properties` | Branch color, branch width, line style of the branches, node dots (size, shape, color, border), highlights, font and size of the tip labels, color of the tip labels, values on branches and nodes, node bars, clade name bars and data plots. `All` and `None` tick or untick them all. | On<br>Off | On | Always |

### › Reset styles

`Reset styles`, at the end of `Page settings`, returns every option of the panel to its default and removes the look given to single elements, as the button of the same name in **Export** does. The tree itself is kept (names, branch lengths, rooting and order, collapsed clades and their names, clade names, fossils marked by hand), and so are the attached files and the reading properties. `Ctrl+Z` undoes it.

## Export

In the **Export** menu of the header.

Save the figure, the tree or a style template.

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `SVG`<br>`PDF` | Vector files that stay sharp at any size. PDF files are compressed, and a figure larger than 200 inches (5.08 m) on a side, the largest page Adobe Acrobat opens whole, is scaled down to fit, still as vectors. | | | |
| `PNG`<br>`TIFF` | Images at the chosen resolution. | | | |
| `Resolution` | | 36 to 2400 dpi | 300 dpi | |
| `Units` | | px<br>cm<br>in | px | |
| `Width`<br>`Height` | Final size of the figure, changing one keeps the proportions. | | Natural size | |
| `Save Newick` | The tree as it is now, with its names, rooting and order. | | | |
| `Save NEXUS` | Same tree, with branch colors for FigTree and its annotations. | | | |

| Option | Description | Values | Default | Appears when |
|---|---|---|---|---|
| `Save the trees (Newick)` and `Save the trees (NEXUS)` | The trees of the sample after the burn-in (and as resampled when read), one tree per line in Newick, or as NEXUS with a note on where they come from. Also in `Export`, Trees of the sample. | | | A file with several trees is open |
| `Save style…` | Saves the look of the figure, without its tree or data. | | | |
| `Apply a style…` | Gives a saved look to the current tree. | | | |
| `Reset styles` | Back to the default look: every option of the panel to its default, and no style of its own on any element. The tree itself is kept (names, branch lengths, rooting and order, collapsed clades and their names, clade names, fossils marked by hand), and so are the attached files and the reading properties. `Ctrl+Z` undoes it. | | | |

# Right-click and canvas

> A `Right-click` sets one element on its own, or the ones selected: a branch, a clade, a tip label, a legend, an image… Values set there win over the left panel, which gives almost complete control over the look of every part of the figure. The canvas is where elements are selected, moved and zoomed.

## Right-click menus

`Right-click` any element (on a Mac: `Control-click`) to edit just that element, or everything selected. Values set here win over the panel, the small ↶ next to a field returns it to the left-panel value. The left panel shows how many elements use their own value, with a button to undo them all.

| Element | What you can change |
|---|---|
| Empty part of the canvas | Tree statistics: tips and nodes, polytomies, rooting, reticulations, branch lengths, ages, fossils and sampled ancestors, tree shape (cherries[1], Colless[2] and Sackin[3] indices, $\gamma$[4]), support values and the tree set it comes from |
| Branch | Color, width, line style, branch length label, grafting a fossil, reroot here, extract clade, make a page, copy Newick, delete |
| Clade | Clade color, highlight background (opacity, start at root, stem or crown, end, gradient, corners), collapse, make a polytomy of the whole clade (and undo it), copy the whole tree to a new tab, choosing in a window whether it keeps the same look or starts from the default look (on the root, where Extract clade does not apply; the copy with the default look keeps what `Reset styles` keeps, and the maps, but not the images placed on the figure), rotate its children (the order of the children of this node), rotate the clade (turned over on its own axis, the order reversed at every node inside it), ladderize, height and fill when collapsed, clade name bar |
| Tip label | Font size, color, style, weight, open nomenclature qualifiers (cf., aff., sp.), extinct tip |
| Node | Dot size, shape, fill, color and border, support label, other node labels, node bars, make a polytomy of its children (and undo it) |
| Grafted fossil | Ages, split the lineage, add fossils to a split, delete |
| Clade name bar | Name, color, sizes, on circular trees, the name along the arc |
| Legend | Title and entry texts, look of that legend, alignment with a selection, hide it |
| Scale bar | Length, unit, sizes, color, alignment |
| Page box | Rows of the page, look of its tree map, alignment |
| Image | Position, size, opacity, rotation, mirror, layer, alignment |
| Label | Text, size, color, bold, italic, typeface, opacity, rotation, mirror, layer, alignment |
| Density plot | Values from (From tree, Tree annotations or Attached files) and Value, logarithmic scale (base 10), cut below and above (where the curve stops, since a density spills past the data), density curve shown or hidden, histogram with its bin width (empty for automatic, in $\log_{10}$ units with the logarithmic scale), color and opacity, smoothing, one color or the colors of the branches or dots (offered only when they are colored by the same values the plot shows, and also applied to the histogram), fill and opacity, line color, width and style, axis with its color, width, text size, spacing and title |
| LTT plot | Count (cumulative or lineages alive), time bins (0 for every event), white background, logarithmic lineages axis, the tree behind with its color, line width and opacity, legend, axis color, width, text size, titles and their spaces to the axes. For each line: its name in the legend, the whole tree or a named clade, fossils included or left out, color, width and style. Lines can be added or removed |

| Element | What you can change |
|---|---|
| Arrow | Color, width, opacity, line style, arrow heads and their size, curvature, space from the anchor at each end, free an anchored end, delete |
| Reticulation | Color, width, arrows and their size, straight or curved line, where it starts and ends along its branches, show its value |
| Value of a reticulation | Text size, color, horizontal and vertical offset |
| Major edge (Full network) | Color, width |
| Value of the major edge | Text size, color, horizontal and vertical offset |
| Any element | Copy style, paste style, reset style |

[1] Cherries: pairs of tips that are each other's closest relatives, hanging from the same node.
[2] Colless index (normalized): imbalance, the difference between the tips on each side added over every node, over its largest possible value, so 0 for a balanced tree (with an even number of tips) and 1 for a ladder-like one. Colless is not shifted by its smallest value, so with an odd number of tips even the most balanced tree is above 0. Needs a fully bifurcating tree.
[3] Sackin index (normalized): another measure of imbalance, the number of nodes between each tip and the root added over every tip, set between 0 for a balanced tree and 1 for a ladder-like one. Needs a fully bifurcating tree.
[4] γ (gamma) statistic (Pybus & Harvey, 2000): when the branching happened along the tree. Below 0, most branching happened early, near the root, and above 0 late, near the tips. Below −1.645, diversification slowed down significantly over time (one-tailed test at 5%). Only for dated, fully bifurcating trees whose tips all reach the present.

## On the canvas

| Action | How |
|---|---|
| Select | Click an element ( `Shift-click` to add more) |
| Select mode | Element (single)<br>Clade ( `Shift-click` extends the selection to the most recent common ancestor)<br>MRCA (pick two tips) |
| Find tips | Type a name or a pattern with * and ? ( `Enter` selects the matches, `Shift+Enter` their clade) |
| Info on hover | Button at the bottom of the panel that shows the tips, branch length and support of the branch under the mouse |
| Move a legend, the scale bar or a page tree map | Hover it and click its lock, then drag it, arrow keys move it 1 px (10 px with `Shift` ) |
| Zoom and move | Mouse wheel to zoom, drag the empty canvas to move, buttons at the bottom right to fit. With more than 3,000 tips, the figure moves as one picture while the gesture goes on, so it stays smooth, and is drawn sharp again when it stops. A tree too tall to fit whole (thousands of tips) opens fitted to the width, from its top, and zoomed far out its lines stay one pixel wide on screen |
| Tabs | Extract clade ( `Right-click` menu of a clade) opens the clade in a new tab, with the same style. Supports, ages and frequencies come from the full tree as they were, and the clade keeps its own branch to show them. Click a tab to switch, double-click it to rename it, the × of the active tab closes it. The tab of the full tree cannot be closed |
| Undo<br>Redo | `Ctrl+Z`<br>`Ctrl+Shift+Z` ( `Cmd+Shift+Z` on a Mac) |
| Clear the selection | `Esc` |